

Substituting Reference Solvent σ -Profiles for Learned Ones Degrades COSMO-SAC Solubility Prediction

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Abstract

Physics-informed models route predictions through physical quantities; accurate references should help. We measure that for solid-liquid solubility: a graph network predicts the charge-density profiles a database tabulates; COSMO-SAC’s fixed residual term and a crystal term turn two into a solubility. Supervising the profile against that database lowers $\ln x_2$ mean absolute error from 2.043 to 1.846 over three seeds; that gain is not certified leak-free, surviving deletion of the scored solutes a leak reaches. Substituting the reference for the learned one raises the error to 2.252 at inference, the training-time rise smaller, at one co-adapted seed. Grounding’s two operations cannot be ordered by size, only by direction. On infinite-dilution activity data another database bounds a deployed thermodynamic model’s error from below, its inputs’ from above; the input bound barely moves, an estimator property, not chemistry, while the error does. One admissibility rule leaves glycol-ether solvents standing, 182 rows over 4 solvents and 19 hydrocarbon and halobenzene solutes, its error exceeding its inputs’ by 2.04 (90% interval $[+1.25, +2.87]$), and 1.76 under the pre-declared estimator. Both are lower bounds; no other subset stands, nor the set whole; solubility admits no such split. That set

is a 59-fold search’s survivor, not an established effect: chemistry-blind relabelling reaches its margin in a tenth of draws under the first bound, a seventh under the other. The profile departs from its reference along a shared direction, on separate three-seed runs. A Hammett pK_a closure reverses the substitution’s sign in distribution, where its constants suffice.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug can be formulated as a tablet or an injection, which solvent a crystallisation is run in, and how a product is purified. It must be measured one solute, one solvent and one temperature at a time, so most industrially relevant combinations have never been determined, and independent determinations of the same system disagree by enough to floor what any predictor can achieve. Dissolving a solid also trades the cost of breaking up a crystal against the cost of mixing the freed molecules into the solvent, so a predictor has to be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph; FastSolv¹ does so with a deep neural network and leads on ex-

trapolation to new solutes. They are the more accurate there and they are opaque: the prediction arrives as a single number, with no account of which part of the chemistry it captured. Physics-informed models instead predict the thermodynamic quantities the physical picture names—an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal—and combine them through a fixed equation, as SolProp² does with a thermodynamic cycle of separately learned components. What that buys is decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data a direct regressor has no way to use. On accuracy the black box leads and the physics-informed cycle trails it. Learned surrogates for the activity coefficient have advanced quickly, and split the same way. Winter et al.³ predict $\ln \gamma_2^\infty$ from SMILES with a language model, and Sanchez-Medina et al.⁴ and Wang et al.⁵ embed a Gibbs–Helmholtz constraint in a graph network for its temperature dependence, all making the activity coefficient the regression target; a second line instead keeps a mechanistic model and learns its input. COSMO-SAC⁶ is a segment-activity reformulation of Klamt’s COSMO-RS, and the σ -profile it consumes—screening-charge density over a molecule’s cavity—is that framework’s construct, tabulated from quantum chemistry.⁷ Predicting it from structure removes that calculation from the pipeline; TeNNet-SAC⁸ does so while also learning the segment-activity kernel and the cavity geometry, and reports accuracy better than COSMO-SAC’s after end-to-end fine-tuning—the opposite headline from this one, on the same broad construction, and §3.3 sets the two side by side.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning,⁹ whose dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,¹⁰ gray-box hybrids composing learned components with fixed mechanistic maps,¹¹ and operator learning with a trainable correction to a misspecified equation.¹² The prevailing expectation

is that more physics improves generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse.¹³ In neural PDE solvers, a learnable term added to a fixed discretised operator absorbs truncation artifacts rather than physics, so the apparent interpretability is illusory¹⁴—though that error source is numerical and has no external reference. The same concern appears in *concept-bottleneck models*: networks routed through a layer of named, human-interpretable concepts, so that the prediction is made from those concepts and nothing else. That is the same shape as a learned intermediate feeding a fixed physical map, and there the failure is called *concept leakage*—a learned concept silently encodes information beyond its nominal meaning.^{15–17} Recent work traces one route to it to the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence the head cannot represent,¹⁸ and overwriting a frozen head’s leaked concepts with ground truth can then degrade accuracy.¹⁹ The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O’Hagan²⁰ and its calibration critique;²¹ four further literatures—forecast decomposition, quasi-maximum likelihood, identifiability, and the validity of composed pipelines—supply pieces used below.

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. For one class of intermediate that can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. It needs testing, because when a predictor of this shape is inaccurate either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies. What this literature lacks is a way to *measure*, rather than diagnose, which of the two limits a composed predictor’s accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after; in the concept-bottleneck case no external reference exists against which either

question could be settled. This work supplies one—a reference σ -profile for solubility, tabulated substituent constants for a second chemical domain—and holds the mechanistic model fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them. The tabulated counterpart can enter at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). Those two entry points are the two operations the word *grounding* names, and here they have opposite signs: training the network toward the reference lowers error, and supplying the reference at the moment of prediction raises it at every seed. Tabulated profiles serve a third purpose on activity-coefficient data, where they divide the error of a fixed model between the model itself and the inputs handed to it. That division cannot be made on solubility, because a solubility measurement constrains the crystal and the mixing contributions jointly and cannot say which of the two is at fault. What it returns is a map rather than a verdict: one answer per stratum, a stratum being a chemically defined set of rows. Whether a fixed model’s own error stands above what its inputs fail to carry is a property of the chemistry the model is asked about, and where it stands above, a more accurate input exposes the model’s bias instead of removing it. The learned distribution meanwhile drifts away from its tabulated counterpart along a direction shared across molecules, and stops reporting the quantity it is named after. Tabulated substituent constants behave the same way inside a fixed Hammett relation, the linear equation that predicts how readily an acid gives up its proton from the constants tabulated for its substituents.

This article is organised as follows. §2 sets out the composed predictor, the identity that divides its error between the fixed model and the inputs, the estimator for each part, the data and the measurement noise that floors what any predictor can reach, and the evaluation proto-

col. §3 reports the signs of the two grounding operations, the division of the error on activity-coefficient data and the chemical map it returns, the drift of the learned distribution away from its reference, the same two measurements on a second fixed model for acid dissociation, where this work’s accuracy stands against published models, and the limitations of all of it. §4 states the design rule the measurements support.

2 Computational methods

2.1 The composed predictor and its error split

For a solute s in solvent v at temperature T , the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades breaking the crystal lattice (Φ) against mixing the freed molecules into the solvent ($\ln \gamma_2$), and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = - \underbrace{\frac{\Delta H_{\text{fus}}}{R} \left(\frac{1}{T} - \frac{1}{T_m} \right)}_{\Phi(T) \text{ (crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with no constant- ΔC_p correction— ΔC_p being the difference in heat capacity between the liquid and the solid. That term is absent from the displayed equation because it is absent from the deployed model, which fixes $\Delta C_p = 0$; the head that would predict it is off in the pinned configuration. The exposure is one-signed and not small. At the group-contribution value this corpus carries, $60 \text{ J mol}^{-1} \text{ K}^{-1}$, the omitted term is a median 0.32 and a mean 0.40 in $\ln x_2$ over the 1318 labelled test rows that carry a measured T_m , 31 solutes, positive on every one of them, which is the size of the substitution penalty of §3.1 and twice the supervision gain. It enters the physics arms and not the closure-free control, and it cancels exactly in the evaluation-time substitution, where Φ is common to the two arms (§S1). What it does not cancel out of is the training of the σ head, which is fitted through a crystal term carrying a bias of that size and in that direction; §3.3 reads the con-

When do reference physical inputs help a learned solubility model?

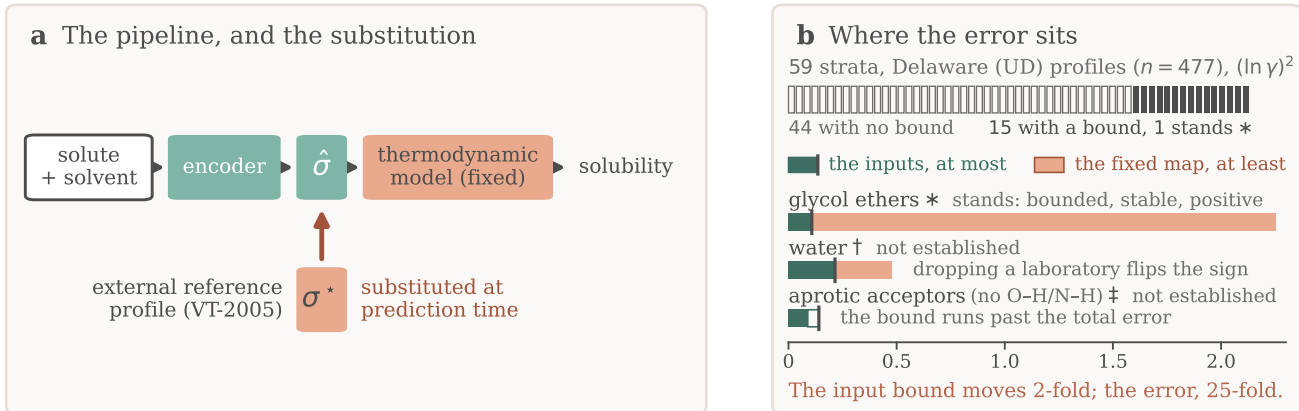


Figure 1: **Overview.** (a) A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility; the arrow marks the substitution of the reference σ^* for it. (b) The counterpart error on infinite-dilution activity data: the strip is every stratum searched, the bars the three solvent classes with a bound; only the starred class stands, each mark carrying its own reason, and each block is a one-sided bound rather than an interval. The right-hand block is the error of the map as deployed, the fixed kernel read at a computed profile, and the profile calculation’s share of it is not bounded here. Training-time supervision is not drawn.

sequence. The activity term is produced by a *differentiable closure* g : a graph encoder predicts a σ -profile—the distribution of screening charge density over a molecule, the histogram a COSMO model consumes—for each molecule, and COSMO-SAC,⁶ a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln \gamma_2$. The segment-interaction energy inside that map—the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other—is the closure’s *kernel*, the part the published revisions rewrite, so a closure variant is named for the year of its kernel: the *2002 kernel* this pipeline runs, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write z^* for the *reference* physical latent, a tabulated σ -profile, and $\hat{z}$ for the head’s own prediction. **Two profile databases supply z^* below and they are never interchangeable:** *VT-2005*, the Virginia Tech 2005 tabulation,⁷ is what the σ head is supervised against and what is substituted for $\hat{z}$ at prediction time, and *UD*, the University of Delaware successor tabulation, is what the activity-coefficient rows of §3.5.1 are matched to. Every set, margin

and figure below names which of the two it is computed on, and no quantity is read across them; the one place they are deliberately set against each other is the database-to-database bracket of §3.5.3. Feeding z^* in place of $\hat{z}$ (a σ -oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^*)$ should be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*; a *DirectGNN* control arm drops the closure and emits $\ln x_2$ directly, and the two were tuned separately (§2.4).

Everything below is stated for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical intermediate $z = h(x)$, against a matched free predictor that shares the encoder and drops g ; solubility is one instance, and a second is the closure of §3.6 for pK_a , the acid dissociation constant on a negative logarithmic scale.

The crystal/activity split of Eq. (1) is not identifiable from solubility alone. At infinite dilution with $\Delta C_p = 0$ the observable is affine in $1/T$, and the activity intercept and slope already span that plane, so the profiled Fisher information for ΔH_{fus} is exactly zero and the indeterminacy set is two-dimensional; at fi-

nite dilution the composition factor $(1 - x_2)^2$ leaves an information of order x_2^2 . The deficiency closes only under a rank-completing external constraint—a direct crystal label, or a fixed activity slope (Lemma S1, §S2). That is why an under-determined factor needs external single-component data and why compensation between the factors is expected; it says nothing by itself about whether grounding helps.

2.1.1 The closure–insufficiency decomposition

Let m be the closure’s target—here the experimental infinite-dilution activity coefficient (IDAC) $\ln \gamma_2^\infty$, in §3.6 the measured pK_a —and $g(z^*)$ the closure evaluated on the reference latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds). *Let g be a fixed (non-fitted) closure and let z^* denote its entire argument (the profile pair together with the temperature wherever the closure reads one). Then (a) the reference-input mean squared error splits exactly,*

$$\underbrace{\mathbb{E}[(m - g(z^*))^2]}_{\text{ref.-input MSE}} = \underbrace{\mathbb{E}[\text{Var}(m \mid z^*)]}_{B_{\text{insuff}} \text{ (irreducible)}} + \underbrace{\mathbb{E}[(\mathbb{E}[m \mid z^*] - g(z^*))^2]}_{B_{\text{closure}} \text{ (decoder bias)}}, \quad (2)$$

the cross term vanishing identically; (b) $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$; (c) $B_{\text{insuff}} \leq \mathbb{E}[\text{Var}(m \mid \text{bin}(g(z^)))]$ for any binning of $g(z^*)$; and (d) B_{insuff} is a functional of the law of (m, z^*) alone and does not involve g , so it is identical across closure conventions, while the bound in (c) is not.*

The proof is in §S2. Two of its parts are conditions of use rather than steps, and both are stated here. **The identity holds because $g(z^*)$ is $\sigma(z^*)$ -measurable, and it fails if any argument of g is left out of the conditioning variable:** with $g = g(z^*, T)$ and T excluded, the cross term need not vanish and the binning is no longer a coarsening of the conditioning σ -algebra, so neither the identity nor the upper bound survives. Where g

reads a temperature—the broad IDAC set of §3.5.1, $\dim = 103$ —we accordingly take $z^* = (\text{profile pair}, T)$; on the VT-2005-matched set every row sits at 298 K and the two readings coincide; and the Hammett closure of §3.6 reads a per-series constant pair as well as the substituent constant, so there $z^* = (s, \sigma_H^*)$ and the conditioning is on both.

B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O’Hagan,^{20,21} and its dominance over B_{insuff} is the empirical content of the claim that the ceiling is set by the closure rather than by the inputs. **What B_{closure} is the error of is a composite object.** The identity compares $g(z^*)$ against $\mathbb{E}[m \mid z^*]$, and z^* is not measured but produced by a procedure—for σ -profiles a quantum-chemical calculation on one conformer and one protonation state per molecule at one level of theory. B_{closure} is therefore a joint property of the map and of the procedure that produced the reference it reads, and not of the kernel’s functional form alone. A displaced reference loads onto B_{closure} through the curvature of g even where g is exactly specified (§S8). The deployed pipeline consumes that same composite, so it is the object a practitioner faces, but every attribution below is to the pair and not to the kernel. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and z^* supplied from outside the fit, so the composite’s error is *attributed* rather than corrected—available only where an external latent oracle exists. **Exactness is contingent on g being fixed:** for a *fitted* decoder the cross term does not vanish and a plug-in point split is invalid. Because B_{insuff} is bounded above and B_{closure} below, a positive *separation margin* $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ establishes that the closure’s own error exceeds what input insufficiency can account for, while a non-positive one is a failure to separate and licenses nothing: a reversal would need a *lower* bound on B_{insuff} , which this instrument does not supply. **Its size is a lower bound and not an estimate.** Since $\text{MSE} = B_{\text{insuff}} + B_{\text{closure}}$ exactly, $B_{\text{closure}} - B_{\text{insuff}} = \text{MSE} - 2B_{\text{insuff}}$, and the reported margin under-states that difference by $2(B_{\text{insuff}}^{\text{up}} - B_{\text{insuff}}) \geq 0$, an amount this design

cannot measure. Every margin below therefore says the closure’s error exceeds its inputs’ by *at least* the number printed, and a bootstrap interval on a margin is an interval on that bound: its upper endpoint limits nothing.

What population, and what sets the resolution. The population is the *measurement superpopulation*: a draw is a nominal compound pair *at a temperature*, together with the conformer, tautomer and protonation state actually populated there, and an independent determination of $\ln \gamma_2^\infty$ for it. Neither the conformational state nor the replicate is part of the conditioning variable, so $\text{Var}(m \mid z^*)$ is the spread of $\ln \gamma_2^\infty$ across the conformational ensemble and across replicate determinations at one reference profile and temperature—non-zero for physical reasons unrelated to sample size. Experimental label noise is a *component* of it and never a term added beside it, and what the design cannot do is *observe* it, neither set holding two rows at a common argument. The bounds are therefore conservative in one direction only—were the superpopulation variance negligible the finding would be the stronger $B_{\text{closure}} = \text{MSE}$ —so nothing reported can be an artifact of over-estimating B_{insuff} , no entry anywhere is the share of the error the inputs contribute, and a low cluster-bootstrap frequency P is loss of resolution and never evidence for the inputs. Resolution is governed by $\dim(z^*)/n$.

2.1.2 The decomposition and the grounding gap

The decomposition is the measurement the conclusions rest on: assumption-light, needing neither a representable encoder nor a lossless bottleneck. Alongside it we report a second, purely observable quantity, the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \leq \mathcal{G} \leq B_{\text{closure}}$, so $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem S1, §S2). A1 is unverified—the linear probe of §S6.1 does not bear on it, scoring an untrained

encoder of the same architecture within its own spread—and A2 is unverifiable and physically dubious; when A2 fails, $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency, a well-specified closure with a lossy latent giving $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$. We therefore read $\mathcal{G}$ as corroborating evidence and base the model-dominance conclusion on the decomposition.

The sign rule, and scope. Whether the *trained* physics model outperforms the *trained* black box at sample size n follows a sign rule.

Remark 1 (Sign of the physics penalty). Write $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$ and $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$, with R_{e2e} and $R_{\text{direct},\infty}$ the two asymptotic risks, $V(n) \downarrow 0$ the estimation variance of each estimator at budget n , so that the physics penalty at budget n is $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$ with $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$ its asymptotic part and $D(n) := V_{\text{direct}}(n) - V_{\text{phys}}(n)$ the variance a constrained prior saves. Then (i) $\mathcal{T}(n) < 0$ (physics helps) if and only if $D(n) > \Delta_\infty$: the variance a constrained prior saves must exceed its asymptotic bias. (ii) If $\Delta_\infty > 0$ and the variance gap vanishes ($D(n) \rightarrow 0$), there is a critical budget n^* beyond which $\mathcal{T}(n) > 0$ for all $n > n^*$.

Both clauses are rearrangements of the definitions above, and this is stated as a remark for that reason: given the two expansions, (i) is an identity and (ii) its limit. What it supplies is vocabulary—a name for the asymptotic part, a name for what a constrained prior saves, and the budget at which they change places—and not a prediction any measurement below is tested against. The limiting refinement in which $\Delta_\infty = B_{\text{closure}} - \mathcal{G}$ when the control is asymptotically Bayes is not a rearrangement and is in §S2, with the working of both clauses. The consequence used below is that a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime. Each physics-versus-control difference reported here is a penalty at the budget actually trained, not an asymptotic one; there is no “optimal grounding” theorem, and the decomposition does not improve out-of-distribution accuracy.

Its variance term $D(n)$ is what carries the semiparametric phenomenon in which an estimated propensity outperforms the known-true one:^{22,23} a fitted nuisance can be the more *efficient* estimator even when its bias is no smaller, so a true-valued input need not give the lower total risk at finite n .

2.2 Data curation: sources, splits, and labels

Solubility measurements come from BigSolDB 2.0:²⁴ 127,930 rows over 15,665 solutes and 221 solvents between 243 and 438 K. The partition is by Bemis–Murcko scaffold—a molecule’s ring systems together with the linkers joining them, side chains removed—into 111,724 training, 8,103 validation and 8,103 test rows, so that every test solute is unseen at training. A melting-point-independent miscibility/structure filter sets row membership, and crystal T_m , ΔH_{fus} , Hansen parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. A split’s rows do not all carry the same label: of the 8,103 test rows, 5608 carry a solubility measurement and the other 2495 carry a melting point and no solubility, so every accuracy number below is on 5608 rows and says so (§2.4). 796 of those 5608 rows, 14.2% of them, carried by 26 of the 147 test solutes, have a solute of more than one component—salts, two ionic liquids and four hydrates—for which the solid in equilibrium with the saturated solution is not the one-component crystal Eq. 1’s ideal term is written for, and not one of those 26 carries a measured melting point. They stay in: dropping the stratum moves the control’s mean and the grounded arm’s the same way and widens the physics penalty this paper reports, so removing it at this stage would flatter the comparison (§S3.2). Solid form is uncontrolled on both label sources—the solubility rows aggregate literature measurements on whatever form each laboratory had, and the $T_m/\Delta H_{\text{fus}}$ labels are merged from a separate compilation that need not refer to the same one—so a polymorph difference enters the ideal term as a systematic and the inter-laboratory disagreement

below as part of its spread; neither document separates it. Crystal supervision is scarce—the test and validation splits contain essentially no measured ΔH_{fus} , the quantitative basis of the identifiability argument (§2.1). The external single-component data used for grounding are an open crystal-property pool ($\sim 15,000$ melting points), the VT-2005 σ -profile database, and ThermoML infinite-dilution activity coefficients.²⁵ Two hazards the pipeline handles explicitly—melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions—are recorded in §S8.

The label-noise floor, and the two functionals it is read with. Independent determinations of the same system disagree, and that disagreement floors what any predictor can achieve. On the raw BigSolDB rows, group by (solute, solvent, temperature to 1 K) and keep the groups reporting at least two distinct sources; within a group the disagreement statistic is the mean absolute deviation of $\ln x_2$ from the group mean. Aggregating that per-group statistic over groups requires a choice of functional, and the two standard ones differ by a factor of five, so both are reported: the mean over groups is $0.15 \ln x_2$ units for the 3948 groups backed by two sources and 0.31 for the 349 backed by three, while the median over the same groups is 0.03 and 0.09. Collapsing exact-duplicate values first—the same measurement re-tabulated across compilations, which would otherwise deflate the floor—leaves the mean at 0.16 over 4261 groups. The scaffold-split MAE reported here (1.70 to 1.85) is above the larger functional by a factor of five, so label noise does not account for the accuracy ceiling under either reading. This floor is a mean absolute deviation in $\ln x_2$; the aleatoric estimate published for this task, an inter-laboratory RMSE of 0.75 in $\log_{10} S$ on 34 duplicate solutions,¹ is a different functional on a different base, and the two are neither pooled nor converted into one another anywhere below.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule

Table 1: Terminology and notation. Each term is defined in the text where it is first used; this table collects those definitions in one place.

Term	Meaning in this manuscript
closure g	the fixed, non-fitted physical map from the intermediate to the observable: here COSMO-SAC, and a Hammett relation in the second chemical domain
kernel	the segment-interaction energy inside COSMO-SAC; a closure variant is named for its kernel’s year (<i>2002</i> , <i>2010</i>)
arm	one trained configuration, or one closure variant evaluated; <i>DirectGNN</i> is the closure-free control arm and <i>TGNNSol</i> the closure-carrying one, abbreviated <i>TGNN</i> in Table 5
z^* , $\hat{z}$	the closure’s <i>entire</i> argument at its reference (tabulated) value, and the network’s own prediction of it; a σ -oracle feeds z^* where $\hat{z}$ would go
grounding	two opposite operations on one reference database: <i>supervising</i> $\hat{z}$ against it during training, and <i>substituting</i> z^* for $\hat{z}$ at evaluation
B_{closure} , B_{insuff}	the error of the closure as deployed—the fixed map read at the tabulated intermediate—and what its inputs cannot resolve about the target: the two terms of Eq. (2)
margin	$\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$, with the MSE of Eq. (2)—the closure read at z^* against its own target m —so in squared m units, squared $\ln \gamma_2^\infty$ on the activity axis and not the solubility axis
binning bound	the upper bound $B_{\text{insuff}}^{\text{up}}$ of Lemma 1(c): the variance of m within bins of $g(z^*)$, which fits nothing to the data
estimator cell	one setting of the binning bound’s four analyst choices—bin count, bin edges, within-bin variance convention, combinatorial convention—together with the row or the pair as unit
$\mathcal{G}$	the grounding gap $R_{\text{orc}} - R_{\text{e2e}}$; distinct from Γ_S , the segment activity coefficient inside COSMO-SAC
stratum, map	a chemically defined row set of the broad IDAC set, and the 59 of them the estimator is run on
boundable	$n \geq 40$, so the fixed cell’s eight bins hold five rows each
admissible	boundable <i>and</i> sign-stable under deletion of each contributing publication, in all four unit $\times$ convention cells
full, residual-only	which part of COSMO-SAC is evaluated, a model-form choice and not an analyst convention: <i>full</i> carries the Staverman–Guggenheim size-and-shape term, <i>residual-only</i> runs the segment-activity part alone and is what this pipeline deploys, its σ head predicting no cavity volume, so the term needs one from elsewhere
VT-2005, UD	the two σ -profile databases, never interchangeable: VT-2005 the Virginia Tech 2005 tabulation the σ head is supervised against and which is substituted for $\hat{z}$ at prediction time, UD the University of Delaware successor the activity-coefficient rows are matched to
labelled test rows	the 5608 of the 8103 scaffold-split test rows carrying a solubility measurement; the other 2495 carry a melting point only and are scorable in no arm
broad IDAC set	477 IDAC $\cap$ UD rows, 185 pairs, all temperatures, UD profiles
VT-2005-matched set	60 pairs at 298 K carrying a VT-2005 profile on both sides
σ	alone, the screening-charge-density profile; $\sigma(\cdot)$ with an argument, the generated σ -algebra; σ_{H} , the Hammett substituent constant, σ_{H}^* its tabulated value and $\hat{\sigma}_{\text{H}}$ the network’s; σ_η , experimental label noise

whose scaffold appears in validation or test. One σ -stream build on disk was nevertheless generated against the wrong split files, and per-run stream provenance was not logged, so the artifacts cannot certify which build fed which arm and the guarantee is a property of the builder rather than of these runs. Of the 147 solutes spanned by the 5608 labelled test rows, 7 carry their own reference profile in that build and 8, those 7 among them, carry a scaffold relative, so the second count is the union and

deleting it deletes both. The ungrounded arm carries no σ stream either, so the supervision contrast is the one a leak can reach and the 0.20 gain of §3.1 is not certified leak-free until the re-run below. The file evidence, the contrast each fact bounds, and the one molecule the leak touches in the $n=44$ analysis of §3.3 are in §S8.

The leak-free re-run. Its protocol is fixed before its numbers exist (§S8): the ungrounded

and the grounded arm at five seeds (42 through 46), the published COSMO-SAC configuration unchanged, on a σ stream rebuilt against *this* split’s files and certified inside each training run, read on the same 5608 labelled test rows and reported seed by seed. So is the decision it settles, which the supervision contrast alone hangs on, stated in three branches and not in the favourable one alone—gains of one sign, gains straddling zero, gains of the opposite sign—and so is the list of displays the five-seed values replace. The substitution result is unaffected in all three branches: it is one checkpoint evaluated two ways, and no σ -stream build can reach it.

2.3 Model, closure, and training

A *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head. Two further backbones are implemented and neither carries a number reported here; the physics-channelled one was run against this encoder and did not beat it, an observation recorded with no set, seed count or interval attached to it (§S1), so no accuracy comparison between backbones is claimed. A fusion head predicts T_m and ΔH_{fus} , giving the ideal term $\Phi(T)$ of Eq. (1). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls: the fitted NRTL (non-random two-liquid) excess-Gibbs-energy model, and its symmetric one-parameter collapse. The third, parameter-free, is the closure under study: a σ -profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$. **That layer runs COSMO-SAC’s residual, segment-activity part alone.** The Staverman–Guggenheim combinatorial term—the contribution molecular size and shape make to the entropy of mixing, and the second factor of $\gamma = \gamma^{\text{res}} \gamma^{\text{SG}}$ in Lin–Sandler—is off in every arm the substitution result is reported on, and the σ head emits a profile shape and a cavity area but no cavity volume, so the term cannot be evaluated from the σ head’s own output.

That is a model-form choice and not an analyst convention: the kernel constants the layer uses are the 2002 values, regressed with the combinatorial term present, so the residual-only setting runs a fitted kernel outside its parameterisation. Its size on a published evaluation is a mean shift of -0.74 in $\ln \gamma_2^\infty$ over 7889 records, from $+0.03$ to -0.71 in bias. The two settings are the *full* versus *residual-only* pair on which §2.5 and §3.5 turn, and one arm of Fig. 2 restores the term by taking a molar volume from a separate auxiliary head and passing it detached, so the combinatorial term is active but the solubility loss cannot train it. Inside the layer, the segment activities are themselves a fixed point, truncated at a count that is a free hyperparameter and is *not* the same in training and scoring: the COSMO-SAC arms here train at 16 segment iterations and are scored at 30. Sixteen was chosen because it lands within 6×10^{-5} of 30 on these weights, so the two agree by construction on this configuration and not as a property of the count; a checkpoint fitted at a different count is scored through an operator it was never fitted against. The exposure that carries to the substitution contrast is not measured. A second, independent truncation sits outside it—the SLE fixed point in x_2 , run to a different count in training than in scoring—and only that one can be re-solved from a deposit: the NRTL arm moves by $+0.0025$ in seed-mean MAE at its training count and a further $+0.017$ to convergence on the outer loop. That figure prices the outer loop and not the segment one. The COSMO-SAC arms can be re-solved on neither, their kernel being a σ -profile pair no deposit and no retained checkpoint carries (§S1). The saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$ as a damped fixed point in x_2 ; a bounded, gated adaptive correction perturbs the physical parameters rather than the output, so the physical decomposition survives it.

External single-component data enter through *grounding streams*—the training-time sense of grounding. Each stream is a sidecar of *self-solvent* rows (no solubility label, only the target factor’s mask active) interleaved into training under the guard above, grounding the

crystal factor from the melting-point pool and the activity factor from the σ -profile database, and so identifying from external data a factor that solubility alone leaves non-identifiable and that a black-box predictor cannot use. Training runs a three-stage curriculum under a weighted loss, and the stages are named by number wherever one of them is referred to below: *P1*, a warm-up on the auxiliary streams alone; *P2*, joint training against solubility; *P3*, a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of *P1*, so the σ stream enters the warm-up only.

The in-house closures, and where their validation stops.

The dispersion-off closures in the fidelity measurement (§3.5.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ -averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer—the *2002, all channels* arm wherever the kernels are compared, so named because it sums the segment contributions of all three profile channels rather than the nonpolar one alone—reproduces the NIST reference COSMO-SAC-2002²⁶ to RMSE $0.0035 \ln \gamma$ on identical VT-2005 profiles. Our COSMO-SAC-2010/dsp layer,^{27,28} built on the *typed* latent of three donor/acceptor-resolved σ -profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off* to RMSE $1.4 \times 10^{-4} \ln \gamma$. That second figure is not deposited, its validator having written no artifact where the 2002 one deposits its comparison, so it cannot be checked against the deposited record (§S1). The dispersion term is *not* validated to that standard, which is why every dispersion-off contrast is computed on our layers while the *2010, dispersion on* runs are quoted from cCOSMO, that reference implementation itself.

The same closure against a published evaluation.

Reproducing a reference implementation fixes the code; it does not say whether the error this paper decomposes is the error a published COSMO-SAC has, or something peculiar to this data assembly. Table 2 answers that. De Souza et al.²⁹ score

the original Lin–Sandler COSMO-SAC on the infinite-dilution activity-coefficient database distributed with *The Properties of Gases and Liquids*, at $\text{AAD} = \mathbb{E}|\ln \gamma_{\text{calc}}^{\infty} - \ln \gamma_{\text{exp}}^{\infty}|$ of 1.75 with HF-TZVP profiles and 0.68 with BP-TZVPD-FINE: one closure, two profile databases, a factor of 2.6 between them. An evaluation of COSMO-SAC is in that sense an evaluation of the profiles it was handed, which is why B_{closure} is defined above on the composite of map and profile calculation and not on the kernel. We ran our own differentiable layer over the same records (§S8), and on AAD it lands inside the bracket the published closure spans across profile databases, under both conventions; on R^2 it is inside in the residual-only convention and level with the bracket’s lower end in the full one, the two printing alike at 0.88. This paper’s own two sets are inside on AAD alone, and in the residual-only convention only—their R^2 , 0.61 and 0.40, is below the published range. Two limits stay open: the profile database is the one axis we cannot match, VT-2005 being what this paper deploys and the axis that publication shows dominates; and the publication’s 6977 points are not proven to be the same rows as the 7889 scored here, the record selection not being stated file by file. Subject to those, the closure error decomposed in §3.5 is, in AAD, the error a published COSMO-SAC has; which half of this paper’s chemical localisation carries across to that set, on 130 times the data and someone else’s compilation, and which half does not, is in §3.5.3. The same publication scores a later kernel on the same records, and its rows are in the table because they put the deployed one in its place: COSMO-SAC-HB2, which types segments as donors or acceptors, reaches AAD 0.46 to 0.48 at R^2 0.96 to 0.97, about half the deployed closure’s 0.77 and a third of the 1.75 that anchors the profile-database bracket. This pipeline runs the 2002 kernel because it is the one whose differentiable re-implementation can be checked against a public reference here, and the results below are substitutions inside a fixed map rather than claims that the map is the best available. §3.5.3 runs the 2002 $\rightarrow$ 2010 step on this paper’s own sets and finds the improve-

ment confined to the smaller set’s chemistry. Every ceiling this paper reports is accordingly this kernel’s ceiling and not COSMO-SAC’s.

2.4 Evaluation protocol, and the DirectGNN control

The *oracle substitution* replaces the predicted σ -profile $\hat{z}$ with the reference VT-2005 profile z^* before the closure, matched by canonical SMILES. What is replaced is one object and not two: the σ -head emits a normalised 51-bin shape and a cavity area whose product is the area-weighted profile the closure reads, and for a matched molecule that pair is replaced by the tabulated pair, which stands in the same relation—so the reference area reaches the residual term’s A_2/a_{eff} prefactor as part of replacing that one object rather than as a separate choice.

A property of the head bears on what that substitution does, and it is a defect rather than a choice. The encoder carries a separate learned adapter for the solute and the solvent role, and the σ -head reads whichever the molecule occupies, so one molecule has *two* predicted profiles. A σ -profile is a property of a molecule alone, and the closure reads the pair, so the model can and does violate an identity no activity model may violate: on a self-pair, where $\ln \gamma_2$ must be exactly zero, the deployed model returns a median -0.058 , a mean magnitude of 0.279 and a maximum of 1.31 over 40 solutes, nonzero beyond 0.01 on 95% of them. With one profile on both sides the same computation returns zero to machine precision, so the violation is the role gap and nothing else. The substitution overwrites both roles at once and therefore also closes that gap, which is a second thing it changes besides the profile values; the two are not separated here. Tying the two roles is the change this calls for, and it is a retraining. The departure this paper reports between the two, $36 \pm 1\%$ and $51 \pm 2\%$ in §3.3, is a relative L_2 distance between *shapes*, both sides normalised, and prices no part of the area. How far the deployed arms’ areas sit from the tabulated ones is not recoverable from the retained artifacts, which keep none of their weights; on the

one trained COSMO-SAC head the tree keeps, the learned cavity area is a median 1.01 times the tabulated one over the solutes it scores and 1.58 times it over the solvents. That head reproduces neither the size nor the direction of the substitution penalty reported here, so those two ratios price the quantity and not the arms. Solute and solvent are matched independently. Nothing else is replaced, the crystal head reading a solute-only embedding the substitution does not reach; the bounded correction does read the closure’s own parameters, so the crystal values the solver finally receives shift by a little: pooled over the three seeds and over the 8066 of the 8103 test rows where the substitution applied, at most 7×10^{-4} in Φ , against a median $|\Phi|$ of 4.5 on that same pooled set. Restricted to the 5608 labelled rows every solubility figure below is scored on, that maximum is 4.2×10^{-4} . §S1 lists every replaced and unreplaced quantity, and gives what the match rule reaches: the reference table covers 99.3% of the labelled test rows on the solvent side and 5.3% on the solute side, so the substitution contrast below is carried by the solvent profile.

It is applied three ways to rule out implementation artifacts: at evaluation only; injected at training time so the model co-adapts; and as a channel-swap control, injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so that only the activity branch refits against it. The three differ in when the replacement is made, not in what is replaced. The last two are at seed 42 only. An analogous override substitutes reference crystal targets. The activity target for the decomposition is the infinite-dilution $\ln \gamma_2^\infty = \lim_{x_2 \rightarrow 0} \ln \gamma_2$, on two sets that differ in their reference database and their regime (§3.5.1–§3.5.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver. The two arms therefore share their representation but not their inputs or their tuning: the control receives an explicit temperature encoding the physics arm does not, seeing T instead through $\Phi(T)$ and the closure,

Table 2: The deployed closure against a published evaluation of the same closure, and this paper’s own two sets through it. AAD is the mean absolute error in $\ln \gamma^\infty$, n the number of scored records.

COSMO-SAC-2002 on	profiles	n	AAD	R^2
<i>As published</i> ²⁹				
PGL 6th ed. IDAC set	HF-TZVP	6977	1.75	0.88
PGL 6th ed. IDAC set	BP-TZVPD-FINE	6977	0.68	0.96
<i>A later kernel on the same evaluation</i> ²⁹ : COSMO-SAC-HB2 ^b				
PGL 6th ed. IDAC set	HF-TZVP	6977	0.48	0.96
PGL 6th ed. IDAC set	BP-TZVP	6977	0.48	0.97
PGL 6th ed. IDAC set	BP-TZVPD-FINE	6977	0.46	0.97
<i>The same set, through this paper’s closure</i>				
full convention ^a	VT-2005	7889	0.98	0.88
residual-only	VT-2005	7889	0.77	0.92
non-aqueous rows	VT-2005	4347	0.47	0.68
aqueous rows	VT-2005	3542	1.13	0.83
<i>This paper’s own sets, residual-only convention</i>				
broad IDAC set	UD	477	1.12	0.61
VT-2005-matched set	VT-2005	60	0.81	0.40

^a The like-for-like row, a published COSMO-SAC carrying its Staverman–Guggenheim term; the residual row is the setting this paper’s B_{closure} is measured in.

^b A donor/acceptor-typed hydrogen-bonding kernel, not the kernel this paper deploys. Printed because it is on the same 6977 records and beats the deployed closure by a factor of about two.

and the optimisation settings were tuned per arm and differ by up to an order of magnitude. The TGNNSolv–DirectGNN gap is accordingly a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, not an isolation of the bottleneck, and every physics-penalty comparison below carries that caveat (§3.7).

The controlled comparisons of §3.1 are run at three seeds and reported as mean±sd on a fixed row set; 5608 is that lock—the test rows that carry a solubility measurement, 5608 of the 8103 (§2.2)—so all arms score identical rows. The 2495 rows outside it carry a melting point and no solubility and are unscorable in every arm alike, the control included; none is dropped for anything an arm predicted. The set is fixed by which rows carry a label and not by which rows an arm handled—the lock’s finite-prediction clause is inert, removing no row from any arm at any seed—and §S3.2 gives the decomposition. The runs behind every number reported here fall into twelve families, a family being a set of runs sharing a script, a corpus and a seed policy; Table S25 gives each number its family, its seed count and the artifact it

regenerates from.

2.5 Estimating the split: one cell, one convention, one rule

The instrument this section fixes is a procedure, and it takes three inputs and nothing else: a fixed closure g , a table of reference values z^* its argument can be read at, and measurements m of its target on rows where both exist. It returns, per chemically defined row set, the closure’s own error MSE, an upper bound on what the inputs cannot resolve, the margin between them, a cluster interval on that margin, and a verdict of admissible or not with the test that decided it. Four analyst choices set the binning bound of Lemma 1(c)—the bin count, where the equal-count bin edges fall, the within-bin variance convention, and the combinatorial convention—and all four are fixed here, before any margin is reported. One setting of them, together with the choice of the row or the molecule pair as the unit, is an *estimator cell*. The four rules below are the procedure in full; the map of §3.5.1 is what running it on this closure returns.

The estimator cell. Eight equal-count bins of $g(z^*)$, the *unbiased* (Bessel-corrected) within-bin variance, and the row unit, applied identically to every set and every stratum whatever its size. The bin count is the choice with the most leverage—it takes the broad set’s aggregate margin from +0.01 to +0.96—so the cell is not left standing alone anywhere a finding rests on it: the same sweep is run for the strata the admissibility rule passes, and reported with them. Enumerating every equal-count contiguous partition, the bin-edge choice is immaterial on the broad IDAC set and material on the VT-2005-matched set, whose margin changes sign inside the envelope; we accordingly read that set’s margin as consistent with zero and carry the claim on the broad set in stratified form.

The convention rule. The margin depends on the combinatorial convention, so one rule governs both sets: every margin is read in the *residual-only* convention, pairing MSE and the bound computed under that same convention. Taking the residual-only convention on the smaller set and the full convention on the broad one would be the favourable choice on each set taken separately; one rule applied to both costs the broad set’s margin about half its value. The Jensen constant-offset bound is convention-specific—it separates under the full convention and inverts under the residual-only default—so what is reported rests on the binning bound, which fits nothing and so lets no held-out row borrow strength from a near-twin. Both bounds are valid on the one population quantity, by Lemma 1(d), and so is the smaller.

The stratification rule. The broad set is stratifiable where the VT-2005-matched set’s 60 pairs are not, on two axes that are pure functions of molecular structure: an ordered nine-class cascade of substructure tests over the solvent, hydrogen-bonding role first and dominant functional group second, and the solute’s role as water or as an organic. Every class name is a substructure test and not a chemical judgement: a *glycol ether* here is a solvent carrying a hydroxyl group and two or more oxygens, and an *aprotic* solvent one with no O–H or N–H hydrogen to donate. The classifier reads a SMILES string and nothing else—never m , g ,

the squared error, the source or the bootstrap—so no cell can have been chosen on its outcome, which is a property of the code rather than an assurance. A stratum is *boundable* at $n \geq 40$, where eight bins hold five rows; the smaller ones are reported at that same cell, with the adaptive bin count $\max(3, \lfloor n/10 \rfloor)$ in the rows n it is applied to carried beside them in the deposited table, nothing being dropped for being small.

The admissibility rule. A stratum is *admissible* only if it is (a) boundable at that fixed cell and (b) robust to leave-one-source-out—its margin keeps its sign when each source publication contributing to it is deleted in turn—in all four unit $\times$ convention cells, since the map reports all four. That last clause is load-bearing and is never dropped: without it the set taken whole passes too. A stratum drawing on one publication fails (b) by non-testability, there being no deletion to perform; a deletion leaving fewer than 16 rows leaves the fixed cell undefined, and fails (b) as well. A third clause reduces the admissible *cells* to row sets, and it is stated here with the same precision rather than left to prose, because the one claim this paper carries from the map and the multiplicity null that tests that claim both turn on it. (c) Admissible cells holding identical rows count once; and an admissible positive row set X is *demoted* to another, Y , when Y is admissible and positive, Y holds more rows than X , more than half of X ’s rows are Y ’s, and the residual $X \setminus Y$ is not itself admissible—so that X ’s margin does not stand outside Y . What decides it is that residual and not the overlap: a row set most of whose rows lie inside a larger one survives (c) whenever its own margin stands outside that larger one, so (c) is not containment and no cell is demoted for overlapping (§S3.8). Clause (c) takes the six admissible cells of §3.5.1 to three distinct row sets; positivity then removes one of the three and (c)’s residual test the second, leaving one. It is rebuilt on every draw of the multiplicity null, on the permuted labels, like the rest of the rule (§S3.6). Everything else is *reported in the map and not stated*, with its n , its sources, its margin, its interval and the reason. The rule governs the cells that favour us and those that do not alike, and it tests whether

a number is stable, not what it means: an admissible stratum whose margin is not positive, and one that (c) demotes, both establish nothing.

3 Results and discussion

A cluster-bootstrap frequency P is the frequency, over two-way (solute $\times$ solvent) cluster resamples, with which the margin it is attached to stays positive; it is not a p -value against any null. Such frequencies are quoted as bands—rounded to one decimal, with > 0.95 and < 0.2 closing the ends—because with 17 solvent and 41 solute clusters on the VT-2005-matched set they do not carry two-decimal resolution; the two-decimal values are in the deposited artifacts. The resampling intervals of this paper—the ranking and recalibration intervals of §3.2, the margins of §3.5 and the pKa intervals of §3.6—are *percentile* intervals: their endpoints are order statistics of a resampling distribution, with no bias correction and no studentisation, over the unit named beside each. §3.2 resamples the 107 solute clusters, 4000 draws, at the 95% level; §3.5 resamples solute and solvent clusters two-way, 3000 draws; in §3.6 the decomposition bounds resample molecules within each Hammett reaction series and the flip gap resamples the four series themselves, 4000 draws. Not every $\pm$ or bracketed span below is one of those. Two in particular are not: the kernel-transfer figures of §3.5.3 are a standard *error* over five fit seeds, and the leave-one-publication-out span of §3.5.1 is the range of that curve. Every claim below carries its set, its size and its scope in the sentence that makes it; the same nineteen are collected side by side as one table in §S3.1 (Table S2).

3.1 Supervising the intermediate toward the reference, and substituting it at prediction time

The word “grounding” names two opposite operations on the same VT-2005 database. Every solubility figure in this subsection and the

next is an $\ln x_2$ MAE over one row set, the 5608 of the 8103 scaffold-split test rows that carry a solubility measurement (Table 1); the single-seed control arms below carry that identical set, so nothing here is compared across row sets. Supervising the σ head against the database *during training* improves solubility MAE from 2.043 ± 0.040 (ungrounded) to 1.846 ± 0.053 (grounded), a gain of 0.20 over three seeds, the two arms’ per-seed values not overlapping. The two arms are not budget-matched: the grounded arm precedes the shared curriculum with a σ warm-up on the grounding pool of up to 40 epochs, early-stopped on a held-out tenth, and the ungrounded arm sets that budget to zero (§S3.2). No arm runs that warm-up against permuted σ targets, so what the gain prices is the intervention entire—extra pretraining and accurate targets together—and not the accuracy of the reference values on its own. Those spreads are over initialisation and batch ordering and price nothing about which solutes fell in the test set; pooling the three seeds and resampling the 147 test solute clusters puts the gain at $+0.198$, 95% percentile interval $[+0.06, +0.33]$. The ungrounded arm carries no σ stream, so this is the one contrast in the paper that the uncertified stream build of §2.2 could inflate rather than one the arguments there bound (§S8). Deleting every solute that build can reach—the 8 of the 147 carrying their own reference profile or a scaffold relative in it, on 351 of the 5608 rows—leaves 2.036 against 1.864 on the remaining 139 solute clusters, a gain of $+0.172$ $[+0.03, +0.31]$. That bounds where the leak can act and does not certify the build: it removes the rows a leaked profile scores, not the effect a leaked profile had on the weights. Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive R^2 (Fig. 2): the model predicts solubility better *from its own learned σ -profile* than when *given the external reference one* (MAE 1.85 vs 2.25, in $\ln x_2$ units, over the same three seeds and with no overlap between the two arms’ per-seed values either). Resampling the same 147 solute clusters over the pooled three seeds puts

that penalty at +0.406, 95% percentile interval [+0.29, +0.53]; the median per-row gap is +0.32 against the mean’s +0.41, so the effect is present at the middle of the distribution and heavier above it. Of the figure’s eight arms two are non-COSMO controls, DirectGNN and a fitted NRTL closure; five COSMO-SAC arms differ in how the σ head is trained—the warm-up asymmetry above included—and in what it is fed, and the sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. σ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, *ref. σ (train)* the co-adaptation control injecting it during training instead.

That +0.41 is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. The control that removes that component injects the reference profile *at training*, so the model co-adapts; at seed 42 it costs +0.18 (1.803 $\rightarrow$ 1.981) where the evaluation-only substitution costs +0.43 at that same seed (1.803 $\rightarrow$ 2.232), so part of the +0.41 goes with co-adaptation and the co-adapted arm is by design free of *that* confound. That part is 0.251 of the seed-42 gap of 0.428, a share of 0.59, and we report it as a description of that seed rather than as a magnitude: the injected arm is what was measured once, and holding it at that value while the learned and substituted arms take their seed-43 and seed-44 values—which moves the numerator too, to 0.301 and 0.261—gives 0.83 and 0.61 (§S3). A channel-swap control costs +0.27, to 2.08. It carries a second confound the co-adapted arm carries too: neither is certified to have trained on the same σ -stream build as the seed-42 comparator both are read against, and the file times settle nothing either way (§S8). Both are worse than the learned- σ arm, which rules out an implementation artifact. Only the three-seed arms can be set against one another, and there the evaluation-only substitution costs about twice what supervision gains (+0.41 against −0.20). That ratio is between two three-seed arms and carries the co-adaptation confound with it, so it

is not the paradox’s co-adaptation-free magnitude; the arm that is, at +0.18, was run at one seed on an uncertified stream build and cannot be ordered against the −0.20 at all, whose own per-seed values (0.237, 0.075, 0.281) straddle it. Which of the two operations is larger is therefore unresolved, and only their signs are carried forward.

3.2 What the substitution costs: level, spread, and the choice of solvent

Figure 3 sets the same rows out predicted against measured, for the control and three of the closure arms. Two things are visible there that the arm ordering is not. First, every arm is severely shrunk toward the corpus mean. The ordinary least-squares slope of predicted on measured is 0.515 for DirectGNN, 0.416 ungrounded, 0.528 grounded and 0.690 with the reference profile, none of them near 1, and in every panel the median trace crosses the identity: sparingly soluble compounds are called more soluble than they are and highly soluble ones less. All four slopes are seed 42’s, the figure’s seed, with no replicate behind them. That shrinkage is not the closure’s doing: at that seed the closure-free control’s slope differs from the grounded closure’s by 0.013, so whatever produces it lives in the encoder and the data. Second, the reference-profile arm is the *least* shrunk and the worst. Its slope is the closest of the four to 1 and its prediction spread exceeds the measurement’s, yet its MAE is the highest and its R^2 is not distinguishable from zero; substituting the reference restores spread that is not aligned with the truth, and its worst region is the sparingly-soluble tail it appears to reach. That tail is where the closure is driven hardest: at seed 42 the substituted arm returns 174 rows with $\ln \gamma_2 > 10$ against none in the learned arm, whose largest is 9.70 against the substituted arm’s 19.5, and 179 of its $\ln x_2$ predictions fall below −15, to a minimum of −22.3 where the corpus itself stops at −17.7. The worst 500 of the 5608 rows carry 0.32 of that seed’s +0.43 mean gap. Neither arm’s fixed point is re-solved

The grounding paradox: reference σ -profiles give the worst arm

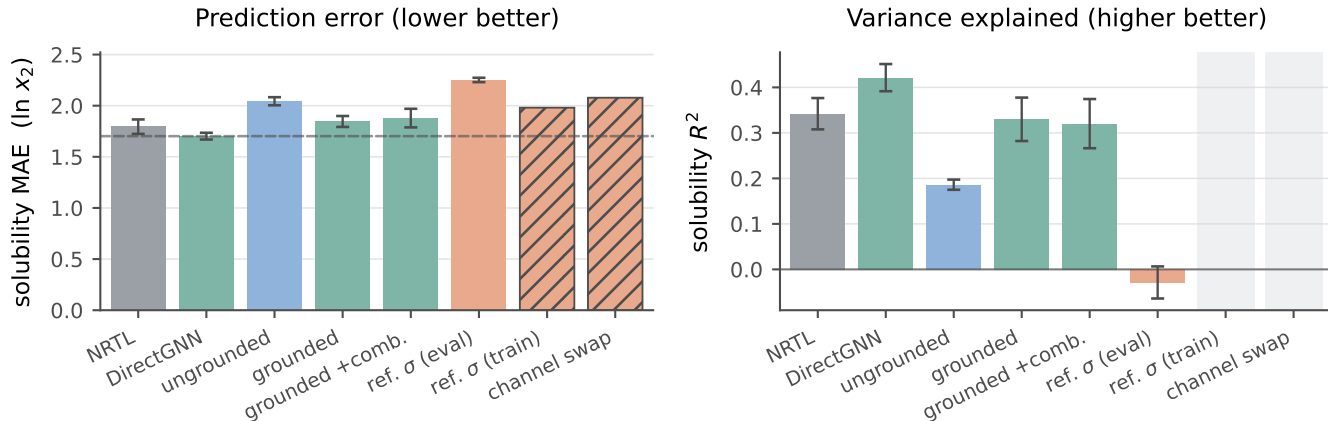


Figure 2: Controlled comparison on the 5608 labelled test rows of the scaffold split. Solid bars: three seeds, mean $\pm$ a population standard deviation over them. The two hatched bars are the co-adaptation controls, seed 42 only, so they carry no error bar and no R^2 , and keep an empty slot in the right panel. Dashed in the left panel only, the DirectGNN reference.

to convergence there, both being scored at the 30 segment iterations of §2.3, so those rows are the ones on which the truncation is least certified as well as the ones on which the two profiles differ most. What the count is worth on these arms is not measurable from the retained artifacts, which keep none of their weights. On the one trained COSMO-SAC head the tree does keep—a retired run whose weights were fit at 8 iterations and which scores 2.61 at 30 against 1.92 at 8 on these same labelled rows—the substitution penalty moves with the count as well: substituting the reference profile costs +4.32 [+3.61, +5.25] at 8 against +6.43 [+5.39, +7.68] at 30 over 5571 rows in 64 solvent clusters whose solvent carries one. That head reproduces neither the size nor the direction of the penalty reported here, its own being fifteen times larger at the count the +0.41 is measured at, and moving its predictions the other way, so those two figures price the lever and not this paper’s +0.41. That is the misspecified map made visible on the solubility axis.

The substitution also costs the decision the model is used for. The separate-tuning confound of §3.7 does not stand in the way here: the reference-profile arm re-runs the grounded checkpoint’s own weights over the same rows and changes only the profile fed to

the closure, so no second tuning stands between the two sides. They differ in Φ by at most 4.2×10^{-4} over these 5608 rows at the worst of the three seeds. The confound it does not remove is co-adaptation: like the MAE gap above this is an input swapped at evaluation, no arm that had trained on the reference profile was ranked, and what co-adaptation would do to an ordering is unmeasured here. Group the labelled test rows by solute and temperature and rank the solvents inside each group (823 groups over 107 solutes; Table S19). The mean per-solute Spearman correlation then falls from 0.592/0.546/0.598 to 0.365/0.369/0.410 across the three seeds, pooled -0.197 $[-0.282, -0.117]$, every per-seed interval excluding zero. Each of the three falls is larger than any difference between seed replicates of a single arm on this measure, the largest of which is 0.060; the deposit carries that benchmark for the rank correlation only, so it is not read against the two measures below. Every interval here is a 95% percentile bootstrap over those 107 solute clusters, the effective sample rather than the 823 groups. Normalised discounted cumulative gain over the top three solvents (nDCG@3) falls by -0.084 $[-0.124, -0.043]$, and top-1 accuracy by -0.107 $[-0.192, -0.020]$. That last figure is quoted

pooled and only pooled: two of its three per-seed intervals span zero, so it is the one of the three ranking metrics that does not hold seed by seed. The solvent the model would select changes in 40 to 53% of groups, asymmetrically: the learned profile alone picks the measured best solvent in 20.0/23.2/21.5% of groups, the reference profile alone in 6.2/14.5/12.2%. None of that is a level. Fitting a separate affine recalibration inside every group—a slope as well as an intercept, the largest concession a reading on which the substitution merely miscalibrates can ask for—removes 86% of the MAE gap, +0.431 [+0.31, +0.56] to +0.060 [+0.02, +0.10], which is the restored, misaligned spread of the paragraph above in another currency. That map is fitted on the rows it is then scored on, two free parameters against a median of five solvents, so it bounds what the concession can buy rather than measuring it. Refitted leaving out the row it scores, it removes none of the gap: +1.09 [+0.3, +2.2] pooled. On the 550 groups holding at least five solvents, where a refit has something to work with, it gives +0.28/+0.07/+0.04 by seed, two of the three intervals spanning zero (§S6.1). The MAE penalty is level and scale in the sense that a per-group map fitted on the answer absorbs most of it; out of sample nothing that sharp can be said in either direction. The ranking penalty is not, under either fit, and by invariance rather than by any centring argument—though the invariance is to an *increasing* affine map, and the fitted map decreases in 10 to 12% of learned-profile groups and 17 to 19% of reference-profile ones, where it reverses that group’s order outright. Restricted to the 589 groups of 823 whose map increases in both arms at all three seeds, the ranking gap is ρ -0.148 [-0.218, -0.079] and nDCG@3 -0.068 [-0.110, -0.026], with top-1 then spanning zero.

What stands in its place cannot be told apart from a solute-blind ordering. Two references know nothing about the solute: the same arm’s own predictions for a randomly drawn other solute over the same solvents, and a ranking of those solvents by their leave-one-solute-out mean measured solubility. Set against both, the learned-profile arm is above them on Spear-

man, Kendall, top-1 and nDCG@3 at all three seeds, twenty-four intervals of twenty-four excluding zero. The reference-profile arm’s own twenty-four margins are small and all but one positive, and it is their intervals that carry the reading: twenty-four of the twenty-four include zero, or twenty-three, one cell sitting on the boundary at [-0.0008, +0.0762] and changing side with the bootstrap draw (Table S19). What that measures is a failure to separate the reference-profile arm from orderings built without the solute, not a demonstration that it cannot be separated from them. Neither arm is chance, the within-group permutation floor being $\rho = 0.00 \pm 0.02$ for both. Supplying the true physical intermediate does not leave the model miscalibrated about solutes; it leaves its ordering of them indistinguishable from one drawn without them.

3.3 What moved inside the model, and whether a downstream correction repairs it

A candidate *mechanism* for the low-fidelity, end-to-end regime is this: a σ -profile head trained *end-to-end* through a fixed, misspecified closure—with no σ supervision in that phase—is under pressure to *cancel* the closure’s error rather than to be physically faithful. The hypothesis has two parts and this section establishes only the first. Its antecedent—that the latent leaves the reference along a direction shared across molecules rather than as per-molecule noise—is measured. Its consequent—that the departure cancels part of B_{closure} —is not, and the one direct check available argues against it. The external reference that fixes the ceiling makes the first part testable: we compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile molecule by molecule ($n=44$). That comparison is the anatomy of the object the two operations of §3.1 act on—how far the learned intermediate has travelled from the reference that supervises it and then replaces it—and it is placed here, inside the phenomenon, for that reason and not as evidence for why either operation has the sign it

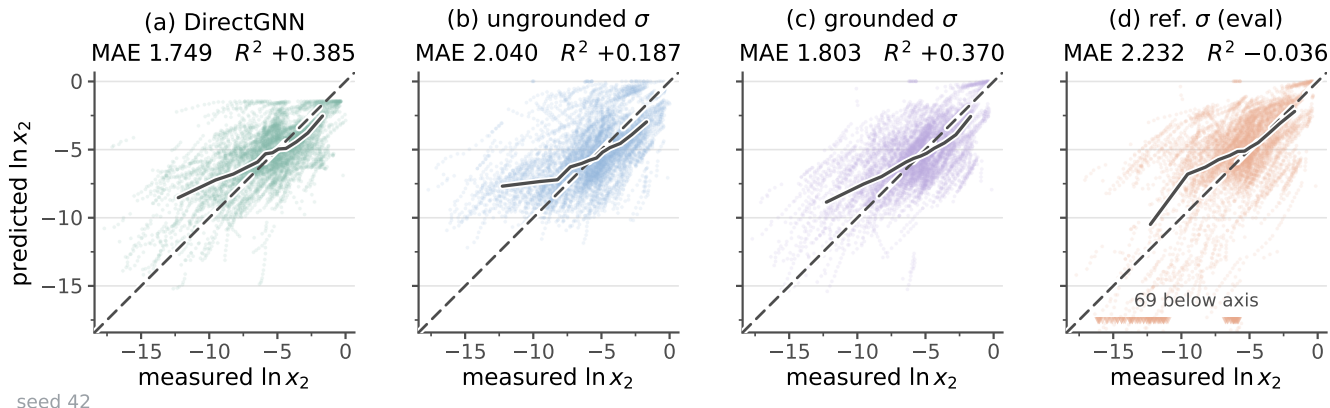


Figure 3: Solubility parity for four arms of the controlled comparison: predicted against measured $\ln x_2$ on the 5608 labelled test rows of the scaffold split, which are 763 solute-solvent pairs over 147 solutes and 70 solvents measured between 248 and 368 K. (a) DirectGNN, the closure-free control; (b) COSMO-SAC over an unsupervised σ -profile; (c) the same closure with σ -profile supervision during training; (d) the arm of (c) with the reference VT-2005 profile substituted at evaluation. Dashed, the identity; solid, the median prediction in twelve equal-count bins of the measurement. Each panel header carries that panel’s own MAE and R^2 . Both axes span the measured range with a margin around it; the markers along the foot of (d) are its 69 predictions that fall below the frame, each at its own measured value.

does.

What is measured, and on what. The comparison is within a single grounded model, checkpointed before and after solubility fine-tuning, over the $n=44$ matched molecules and repeated at three seeds. Those three runs are *not* the arms that produce the paradox: they are a separate set, trained with the σ head left unfrozen through fine-tuning, where the four COSMO-SAC arms of §3.1 hold the head from P2 onward. That freeze holds the head and not the encoder feeding it, so those arms’ profiles are not pinned at their warm-up value either; how far *their* latent drifted is not measured and cannot be recovered from the deposit. What follows is therefore measured on runs of its own and is not established of the checkpoints whose paradox it is offered to explain. Under σ supervision the head reproduces the reference profile to within $36 \pm 1\%$, so the intermediate *can* carry the physical shape—though only as an in-sample fit, all 44 probe molecules lying inside the σ -grounding pool. Solubility fine-tuning then drives it a further $41 \pm 3\%$ away, for a total departure from the reference of $51 \pm 2\%$ —a magnitude reported but not stable across anal-

ysis configurations, so what these runs fix is the sign and the shared direction below and not the size. A relative deviation carries no scale of its own, and the ladder that supplies one (§S5: the same 44 molecules and the same relative- L_2 metric, its rungs built from the reference profiles the grounded arm was supervised on) places the fine-tuned latent at 0.507 and the grounded base at 0.364, against 0.493 for one corpus-mean profile reused for every molecule—a rung carrying no molecular information at all—and 0.651–0.684 for each molecule given another molecule’s reference profile. Read on that ladder, the drift consumes essentially all the molecular specificity grounding had provided, which is what makes the shrinkage reading below more than a formality.

What the departure is, and what it is not. The drift is not per-molecule noise and its structure is not a constant offset: the first two principal components of the mean-centred deviation carry $72 \pm 1\%$ of the variance, and the quantity read against a null—the top-two share of the learned-versus-reference deviation, 0.399—clears a phase-randomised null (0.221, $p < 0.001$), on one grounded checkpoint rather

than over the three seeds, the 0.221 carrying a spread of ± 0.01 that is the null’s own over draws and not a spread over seeds. Two nulls it does not clear travel with that: the mean-centred share has no matched null of its own, and a wrong-molecule null built from real reference profiles does *not* clear (0.772, $p = 1.00$), so if the case for preferring the phase-randomised null is not accepted the spectrum is uninformative rather than supporting. One distortion operator fitted on half the 44 molecules predicts the drift on the other half $4.1 \pm 0.5\times$ better than the zero-drift null, the $\pm$ spanning the three seeds at one partition of the 44 and not the partitions. That null is a deliberately weak one and is uncontrolled on these three seeds: a molecule-independent constant offset already reaches $2.1\times$ on the one companion run where that control can be computed. So the ratio stands against the identity null alone, and structure *beyond* an offset rests on the mean-centred spectrum. The held-out half is also homologue-saturated—six n -alkanes, four halobenzenes, five pyridines and four butyl and hexyl amines among the 44, and the median held-out molecule sits 16% of its own profile norm from its nearest training-half neighbour—so “outside the fit” is largely within-cluster interpolation. That spectrum does not exclude a shrinkage of every profile toward the corpus-mean shape, which is how the drift looks on the reference ladder of §S5. The dominant mode (Fig. S4) moves area out of the nonpolar bins and into the polar wings: fine-tuning makes the network represent molecules as more polar than its *own grounded profile* does—the plotted drift being taken against that profile and not the reference, and the mode defined only up to sign, drawn in the mean drift’s direction.

Whether the departure cancels closure error. It is not shown to, and the one direct check runs against it. Holding the cavity area at its reference value, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint scores 18.75 against 1.47 on the $n=60$ VT-2005-matched pairs: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure. Four lim-

its sit on that number—the checkpoint is the uncontrolled one, whose 82% departure is half again the three-seed $51 \pm 2\%$; that checkpoint is a retired run, trained at 8 segment iterations and scored at 30, and its own scaffold-split accuracy is MAE 2.61 at $R^2 = -0.31$, worse than every arm of Fig. 2; the axis is wrong, the drift being produced at finite composition and scored at infinite dilution; and the check has almost no power in the direction that matters, so it separates “no cancellation” from “large cancellation” but not from “some cancellation”. The kernel sweep reported beside it does not add a second check in the same direction, and it is stated here for what it does bound. Sweeping the closure’s hydrogen-bond coefficient at zero, one and two times its standard value over those same $n=60$ pairs gives mean squared errors 2.42/2.83/10.15 on the strong-donor pairs, 0.090/0.092/0.918 on the acceptor-only ones and 1.205/0.867/0.578 on the inert ones. The three classes separate by direction rather than by size: both hydrogen-bonding classes rise with the coefficient—the acceptor-only pairs by tenfold and the strong donors by fourfold—while the inert class falls monotonically by half. That is consistent with B_{closure} sitting in the hydrogen-bond kernel and does not localise it there. The classes are assigned by the *solvent* alone, and five of the eight inert-class pairs carry a solute that does have donor-window mass—five primary or secondary amines, 1.2 to $3.8 \text{ \AA}^2$ of it against benzene—whose own pure-component segment activity the coefficient scales at infinite dilution. The coefficient therefore has a route into the class its name says it should not touch. For the other three it has none: 3-methylpyridine and tripropylamine, twice, carry $0.000 \text{ \AA}^2$ there, and the exchange energy is a product of a donor and an acceptor segment, so no scaling of the coefficient reaches them. Those three are unaccounted, and the improvement on the class as a whole is not evidence of a knob acting where it should not. It says nothing about a more polar *profile*, a coefficient not being a profile. On the strong donors the error at $c_{\text{hb}} = 0$ is 2.42 against 2.83 at the standard value, so switching the deployed hydrogen-bonding term off lowers

the error on the class it governs—on 28 pairs, 13 of them carrying one solvent and 20 carrying two, with no interval, on the set §3.5.2 declines to stratify at all. Neither the two-MSE check nor the sweep refutes cancellation in the regime that generates the drift; they sit on the infinite-dilution axis, they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. A second reading is live and is not excluded here. ΔC_p is zero in the deployed ideal term, worth a one-signed median 0.32 in $\ln x_2$ on the test rows carrying a measured melting point (§2.1), and the crystal head is supervised toward measured T_m and ΔH_{fus} and cannot absorb it, so the residual lands in $\ln \gamma_2$ —and a profile pushed toward its polar wings is what compensating an over-large Φ looks like. The design does not separate that from a misspecified activity kernel. What is established is weaker and still consequential: the model’s “physical” intermediate is not the molecule’s charge distribution, consistent with cautions that the interpretability of physics-grounded latents is not guaranteed under misspecification,^{21,30} and that adapting a representation can itself collapse an otherwise physically-faithful latent.³¹ Its *structure*—low-rank rather than diffuse—excludes a random collapse but not the structured shrinkage above. The full measurement, its nulls and the two controls are in §S5.

What leaves σ under-determined by activity, and what freezing it buys. The freedom that would make such a cancellation possible is the closure’s geometry, measured in §S5: at infinite dilution the activity map’s Fisher spectrum is effectively rank one—participation ratio 1.0 in aggregate and ≈ 1.4 of 51 per molecule, the resolved direction lying on the hydrogen-bonding wings of the profile—so the end-to-end head can move σ along the remaining, low-sensitivity coordinates at little cost to the fit—the σ -profile form of Lemma S1. Nothing there shows recovery of the physical profile to be impossible in principle. The sign of grounding turns on *both* axes, and on which sense of grounding is meant. Freezing the σ head at its supervised warm-up state instead of

letting it drift end to end cuts the reference-input penalty by 65% (ΔMAE from +0.40 to +0.14 over $n=5571$ matched test rows). Those are a separate matched pair of runs—one base, at a shorter P1 than the pinned schedule, resumed twice with the freeze off and on, so a single pair carrying no seed replicate, and the 65% is a difference between two runs and not a mean over any—and not the +0.41 arms of §3.1. A near-physical intermediate makes the pipeline far more robust to substituting the reference σ . Supervision alone does not, however, *flip* the sign here—the penalty stays positive, the closure still being the low-fidelity COSMO-SAC-2002 and the warm-up σ itself only partly physical.

The nearest published construction, which reports the opposite sign. TeNNet-SAC⁸ is this pipeline’s closest relative in print: a σ -profile predictor feeding a COSMO-SAC-shaped segment-activity model, with its profile and geometry predictors trained on 39,745 quantum solvation calculations and its segment-activity predictor pretrained on a million synthetic points and then fine-tuned end to end on experimental activity coefficients. It reports accuracy comparable to COSMO-SAC in its base form and, after that fine-tuning, consistently better than COSMO-SAC across its benchmarks. Three differences separate it from what is measured here, and only the last is a hypothesis. The first is which operation is scored. TeNNet-SAC’s gain is a training-time gain from supervising and then fine-tuning a learned profile, and that is the operation this work also finds positive (+0.20, §3.1); the operation this work finds negative is substituting a tabulated profile for the learned one at inference, which TeNNet-SAC does not report. On that reading the two results do not conflict. The second is that it carries a geometry predictor for molecular volume and surface area, so its composed model can evaluate the combinatorial contribution this one cannot (§2.3). The third is that its segment-activity kernel is trainable and this one is fixed at the 2002 parameterisation. If the sign of the substitution turns on whether the kernel can move, the find-

ing here is about a fixed 2002 kernel and not about routing a prediction through a physical intermediate, and §3.4 already records that no grounding gap has been measured under any modern closure. Nothing here measures that: it would take the same two arms retrained against the 2010/dispersion kernel end to end, which is limitation (x)’s named run and was not made. What it would take to reconcile the two through profile fidelity instead is a σ -profile departure on TeNNet-SAC’s side well below the $36 \pm 1\%$ this pipeline reaches under supervision and the $51 \pm 2\%$ it reaches after fine-tuning; that departure is not reported there.

Whether a downstream correction repairs it. If the closure binding were merely an inaccurate point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not: an additive bounded residual with a learned confidence gate leaves the gate effectively shut, and forcing it open removes the systematic *bias* ($+0.45 \rightarrow +0.06$) but recovers little *accuracy* (R^2 $0.236 \rightarrow 0.274$ on the same rows), as expected if the closure, not the estimate, is the binding constraint. Both arms are seed 42 only, and this is a statement about the one correction tested: an additive bounded residual on the output, gated, on this closure and this corpus.

The learned profile is chemically inadmissible in the window the closure’s hydrogen-bond term reads. COSMO-SAC’s hydrogen-bonding contribution acts on segments beyond $|\sigma| = \sigma_{\text{hb}}$, and for a molecule with no hydrogen-bond donor the donor side of that window must be empty. In the reference tabulation it is, exactly: cyclohexane, acetone, toluene and tetrahydrofuran each carry $0.000 \text{ \AA}^2$ of donor-window area, against 6.20 for ethanol and 15.75 for water. The learned profile carries 44.0 , 32.4 , 24.5 and $50.7 \text{ \AA}^2$ in that same window—between 23% and 36% of each molecule’s total surface—and every row of the scored set carries mass on both sides of it, so the hydrogen-bond kernel is active on every pair regardless of chemistry. Those areas are

read off the one trained COSMO-SAC head the tree holds, which is a retired run, trained at 8 segment iterations and scored at 30, and scoring MAE 2.61 at $R^2 = -0.31$ on the scaffold split—worse than every arm of Fig. 2. The null arm that separates architecture from training sits in the same deposit and is reported here because it exists. Measured as a fraction of profile mass in the same extreme bins, where the reference tabulation’s median is 0.000 , an *untrained* network of this architecture reaches 0.212 against the trained head’s 0.189 . The window is full before any training, and training emptied it slightly rather than filling it.

Two causes are separable in principle and are not separated here, and the head this is measured on decides which is live. Its σ -supervision stream ran for zero steps: the profile received no supervision at all, and §S6.2 is the reason that matters—solubility alone does not determine the profile’s shape, so an unsupervised profile is free in exactly the bins that carry no constraint. That accounts for a quantity of this size. The parameterisation accounts for something much smaller and of a different kind: the profile is emitted through a softmax over the 51 bins, which cannot return an exact zero where 57% of the reference table’s bins, over its 1424 profiles, are exactly zero, so no amount of supervision empties the window—but a softmax reaches 10^{-6} at a logit gap of ten, and a floor of that size is four orders below what is measured. The measurement therefore bounds neither cause on its own. What it does establish is that the window is not empty in a trained arm and that the closure’s hydrogen-bond term is consequently live on every pair, which is a defect whichever cause carries it; a supervised arm separates the two, and a head able to emit zeros removes the architectural half.

3.4 What the substitution does not decide, and what would settle it

Which of the two channels the substitution fails through is not decided here. Drug-like solutes are essentially absent from the VT-2005 set, so

the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers only ≈ 7 solutes; enlarging that set would mean computing new reference profiles rather than looking them up.

Nor does anything above decide *why* the substitution hurts. The reading it invites is that handing the closure a faithful input surfaces a misspecified map rather than repairing it, so that the reference profile is the better input only where the map that consumes it is faithful. That reading is untested here, and for a stronger reason than a test that returned nothing: on the solubility axis the two terms of the split do not separate at all, and the solubility axis is where the substitution of §3.1 was measured. The quantities the reading is about—a fixed map’s own error, and what its inputs fail to carry—are therefore not observable on the rows §3.1–§3.3 report.

§3.5 measures a second axis rather than explaining the first. It uses a different reference database, a different error functional, no trained model at all, and a row set that shares none of the rows above; the two are measured in different regimes on largely disjoint chemistry, and no measurement in this paper connects them. Until the connecting experiment is run, “the substitution hurts *because* the map is misspecified” remains untested.

Two experiments would settle it and neither has been run. The first is the transfer back to the paradox’s own regime—finite composition and all γ , where the substitution is measured, rather than infinite dilution, where the split is—which is limitation (i) of §3.7 and the principal open question here. The second is limitation (x), and it has a part the fidelity contrast of §3.5.3 does not supply: that contrast compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself is measured under no modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested. Both runs need a typed σ -profile head trained end to end through the 2010 kernel, and the solubility one additionally needs a $UD \cap \text{BigSolDB}$ matched set, which does not exist here.

3.5 A second axis: bounding a deployed COSMO-SAC against its inputs

The estimator is the binning bound, read at the one estimator cell fixed in advance in §2.5 and used nowhere else in this paper: eight equal-count bins of the conditioning variable, the residual-only convention, the unbiased within-bin variance, and the admissibility clauses (a)–(c) that decide which strata may be read at all. All four were fixed before any margin was computed; none of §3.1–§3.4 is measured with them.

The decomposition of §2.1.1 separates a fixed closure’s error into a part in the map and a part in its inputs. Its output is a map over chemical strata rather than one verdict for a set, for the reason §3.5.1 establishes, and it runs on two data sets that must not be conflated, the *broad IDAC set* and the *VT-2005-matched set* of Table 1.

One reading rule governs every margin below, on either set and under either kernel, and it is stated once here rather than at each of them. The comparison is one-sided in both directions it can be read: a positive margin establishes $B_{\text{closure}} > B_{\text{insuff}}$, while a non-positive one is a failure to separate the two terms and licenses nothing—it is not a measurement that they are equal, and no cell of the map can ever return “better inputs would help here”. The size of a margin is a *lower bound* on $B_{\text{closure}} - B_{\text{insuff}}$ and not an estimate of it (Lemma 1, §2.1.1), so an interval quoted on a margin is an interval on that bound and not on the difference itself. And B_{closure} throughout is the error of the fixed map *read at a computed profile*, so what a positive margin establishes is a property of that composite and not of the kernel alone (§3.5.3). The one estimator cell above serves both sets, both kernels and every stratum, so no cell below was fixed after the number it returned was seen.

3.5.1 The broad IDAC set ($n=477$)

The closure reads a temperature as well as a profile pair and is evaluated at each row’s

own temperature, so the conditioning variable is its full argument (z^*, T): the concatenated UD profile pair with T , 103 dimensions against $n=477$ ($\dim/n \approx 0.22$). The target m is the experimental $\ln \gamma_2^\infty$, so floor and closure error are measured on one data set and one profile database, that database being UD throughout this subsection. Its chemistry is small: no solute exceeds 12 heavy atoms (median 7), against a median of 14 and a maximum of 122 for the drug-like solids of this paper’s own corpus, and only 0.46% of corpus rows carry a solute from this set. *Broad* therefore names the reach of the γ^∞ corpus the kernel is calibrated on and not coverage of the chemistry a solubility model is deployed against: it covers the solvent side of deployment and not the solute side. It is matched to UD by InChIKey with a first-block fallback, looser than the VT-2005 set’s rule. A displaced z^* inflates B_{closure} and the binned $B_{\text{insuff}}^{\text{up}}$ together and its direction on the margin is unsigned; what carries the point is the re-estimate, the aggregate margin being unchanged on the 465 exactly matched rows.

Every MSE, bound and margin on this axis is in squared $\ln \gamma_2^\infty$ units, and none of them is on the $\ln x_2$ axis the MAEs of §3.1 are measured on. Under the convention rule of §2.5 the *2002, all channels* arm scores $\text{MSE} = 1.902$ against $\text{Var}(m) = 4.85$, and the binning bound gives $B_{\text{insuff}} \lesssim 0.70$ and hence $B_{\text{closure}} \gtrsim 1.21$. The separation margin for the set taken whole is $+0.51$, against $+0.95$ under the full convention and $+0.27$ collapsed to one row per distinct pair. It is positive in every cell of the bin-count $\times$ variance $\times$ convention grid, and two of the three bootstrap clusterings extend just below zero.

That aggregate is not the instrument’s output. Two operations reported separately—the pair unit, and dropping the two chemistries the error concentrates in, the glycol-ether solvents and the rows carrying water as the solute—compound badly. The chemistry cut alone takes $+0.51$ to -0.12 on the 258 rows that remain, the pair unit alone gives $+0.27$, and the two together give -0.38 on 136 pairs; under the full convention the same pair of cuts takes $+0.95$ to $+0.28$ to $+0.01$. Nor is the aggregate

carried by the set: deleting one source publication at a time, `10.1021/acs.jced.7b00114`—37 rows, six molecule pairs, 32% of the set’s squared error—takes the margin to $+0.18$, while each of the other fourteen deletions leaves it between $+0.42$ and $+0.87$ (under the full convention that curve is flat, $[+0.82, +1.07]$ against $+0.95$). That single deletion loses two thirds of the margin and its resolution without reversing its sign; the sign reverses only under the compounded cut. Two caveats qualify the corroborating estimators and not the bound—the out-of-fold ones leak under random folds and are uninformative or split under blocked ones, and the tighter binning-free pair bound samples no between-laboratory component and is therefore not promoted—so together they say the binning bound is slack rather than that B_{insuff} is negligible (§S3). What is reported rests on the binning bound (§2.5, Table S9).

The map, and how much of it survives the rule. The instrument runs over 59 strata of the broad IDAC set—477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications—on the solvent axis at three granularities and the solute axis at two, plus their cross-tab and the set as a whole. Fifteen have $n \geq 40$ and are boundable, the only ones whose margins the rule can act on. Table 3 prints seven of them—every one the rule leaves standing and the classes they are read against—with their n , their sources, their margins, their intervals and the condition each is in; the remaining eight are in §S3.8 in the same columns. The other 44, including the 24 that carry no estimate at all, are printed with the test each fails in §S3.8 (Tables S11–S15 and Fig. S2), every stratum with its error, its bound, its interval and its share of the set’s squared error, nothing dropped for being small. No stratum but the one named below is stated as a finding anywhere in this paper. Six of the fifteen are *admissible*. Clause (c) of the rule makes those six three distinct row sets, of which one is both positive and not demoted by its residual test: the glycol-ether solvents. **That row set is the one the admissibility rule leaves standing**, at a margin a chemistry-blind relabelling of the same molecules reaches in 10.1% of draws (be-

low). Its clusters are few and its chemistry is narrow, and both belong in the sentence that states it. The 182 rows over 43 pairs carrying 45% of the set’s squared error span 4 solvents—ethane-1,2-diol and the di-, tri- and tetraethylene glycols, one homologous series—and 19 solutes, every one of them a hydrocarbon or a halobenzene (108 alkane rows, 48 aromatic hydrocarbon, 24 alkene, 2 halogenated aromatic); 175 of the 182 rows are gas chromatography, and two of the three publications carry 177 of them. On those rows the closure’s own error exceeds what input insufficiency can account for by at least +2.04, that bound carrying a 90% interval of [+1.25, +2.87] at the residual-only convention and reaching +2.94 at the pair unit under the full one. It is positive in all four cells at $P > 0.95$ in each, and under every single-publication deletion, the worst leaving +1.60 when 10.1016/j.jct.2016.10.013, which carries 53% of that stratum’s squared error, is removed. The unit the stratum’s chemistry actually varies over is the solvent, and the deletion that unit calls for is not the publication’s: three publications map onto four molecules, so leaving one out is the sharper test. Deleting each glycol in turn leaves +1.72, +1.94, +2.05 and +2.29, and taken one at a time the four give +3.17, +2.44, +1.71 and +1.21, falling monotonically with chain length on 41, 36, 56 and 49 rows. The sign survives every deletion; the magnitude runs over a factor of two and is set by the shortest oligomer. Read the four one-at-a-time values as a shape and not as four bounds: diethylene glycol’s 36 rows are below the $n \geq 40$ the map’s own boundability rule asks of a cell, so that one is printed here for the trend it belongs to and is not admissible on its own. That interval is the two-way cluster bootstrap run to convergence, at 1.2×10^8 draws against the 3000 each cell of the map draws. Its ± 0.02 of Monte-Carlo spread at 3000 is this cell’s own and does not carry to the others: measured per row set over 200 seeds, the spread of an endpoint any seed moves runs 0.0009 to 0.223, and 42 of the 46 endpoints drawn at 3000 stood inside four standard errors of the boundary their second decimal turns on. The other four are not the ones that converged: they are

two cells with one cluster on one of their two axes, so the resample is one-way over three clusters, about ten distinct compositions exist and no seed moves them at all. Each interval in the map therefore prints at the precision its own draw count determines, and the eleven that neither two decimals nor one would reach carry their Monte-Carlo spread beside them (note *b* of Table S11). That rule settles the Monte-Carlo question and not the sampling one, and the two are different: the second decimal of [+1.25, +2.87] is a property of the draw count, while the interval’s solvent axis resamples the same four molecules the deletions above delete, so how far the endpoints would travel over a different four glycols is not a question this bootstrap or any other on these rows can answer. The deletion spread is the honest width on that axis. It is not an artifact of one solute family: the 74 rows left when its alkane solutes are dropped keep the sign at +1.40 residual-only and +1.69 full, under all three deletions. Nor is it a property of the estimator cell. Table S5’s bin-count grid is the analyst choice with the largest leverage here, taking the aggregate from +0.01 to +0.96. Swept over it, the stratum is positive in all 56 row-unit cells and all 40 the pair unit defines, the residual-only unbiased column running +1.06 to +2.19, and the deletion test passes at every one of the 36 cells where it can be run (Table S6). The occupancy rule that fixed the eight-bin cell would ask for 23 bins at $n=182$, so the cell the map reads is the coarser and the lower of the two. That sweep varies the binning bound’s own settings and not the family of estimator. A pre-declared cross-fitted regressor, tested after this analysis was fixed, returns a *looser* bound: it moves this stratum to +1.76 [+1.02, +2.49]—one 3000-draw replicate, whose second decimal is seed and not estimator (§S3.12)—leaving it the one admissible positive row set but with fewer cells around it, and moves the chemistry-blind null the *wrong* way. That figure is at the residual-only convention and row unit the +2.04 is read at. Under the declaration’s conservative fold variant, grouped on the source publication rather than the molecule pair, no admissible positive row set survives at all: this stratum’s cross-fitted

bound there is 4.92 against an MSE of 2.25, for a margin of -7.58 , on 15 source groups split five ways. That bound exceeds the stratum’s own total error, which B_{insuff} itself cannot do, an upper bound on it being free to be arbitrarily loose, and the regressor behind it scores $R^2 = -0.60$ out of fold on those rows: the estimator failed at that fold count, and the margin is uninformative there rather than a reversal—the same reading §3.6 gives its own negative floor. That null is re-run on both sides for that comparison, on the 473 rows the cross-fitted estimator scores rather than the 477 the map is built on, so its baseline column is not the 10.1% and 17% quoted below. Against a binning-bound baseline of 0.094 on those rows the cross-fit gives $p(D) = 0.137$, and the share of certifying draws that reach the observed maximum goes from 16% to 54%—the difference between a margin at the ninetieth percentile of its own search null and one near the median of it. The declaration required the cross-fitted estimator to be adopted UNCONDITIONALLY if it passed a validity gate that reads no margin, and it passed; every margin printed in this paper is nevertheless the binning bound’s. That deviation runs in this paper’s favour, and it is a choice made after the two bounds were compared, which is what a declaration exists to prevent. The reason is that the rule was written against one hazard—taking the smaller bound cell by cell once the cells are seen, which is biased low—and did not anticipate the other, that one valid bound would be uniformly *looser* than the other. Obeying the rule there prints a bound that is valid and strictly less informative: every claim weakens and none becomes more correct. What limits the deviation is that the number it was made after is a single global comparison of two bounds on one row set and not a per-cell selection, and that the poorer cross-fitted map is printed cell by cell in §S3.12, where the declared rule can be held against it. What the sweep establishes is therefore robustness within the binning bound, not across estimators.

What the row set does not have is a multiplicity null it clears, and this belongs beside the finding rather than in a limitation. It is by

construction the survivor of a 59-fold search. The null relabels the chemistry blindly: each solvent and solute keeps its rows and takes another’s class labels, and the stratum family is rebuilt and put through the same rule. Over 2000 such relabellings of the same molecules, on all 477 rows and at the estimator cell the map deploys, the map’s six admissible strata are above chance ($p = 0.01$) and its two admissible positive row sets are not ($p = 0.286$). Such a relabelling certifies at least one row set in 58% of draws. The size is the sharper statistic, and it is not sharp enough: a certified margin of $+2.04$ or more arises in 10.1% of all draws, which is 17% of the draws that certify anything, against a median of $+1.57$ among those. The separation is what the instrument licenses on the rows it was computed on; it is not established as a chemistry-specific effect against the search that found it. Each p here is a frequency and not a bound in either direction, and the reason is two biases in the null that are measured rather than argued. The first is that provenance survives the permutation, publications measuring related sets of solvents: a label-permuted map has *more* boundable cells than the chemical one (62 against 50, median over draws) and each passes the deletion test *less* often (0.56 against 0.70). Those two halves run in opposite directions and are not netted, so the 10.1% in particular cannot be read as conservative. The second runs one way. The set taken whole has a positive margin at this cell, so a randomly assembled stratum inherits a positive margin on average, and a null frequency near one half for certifying something is expected rather than evidence that the rule is loose; that bias inflates the 58% and not the 10.1%. The exactly matched null would permute chemistry and provenance jointly, which these data do not allow, the two being confounded in them (§S3.6).

What settles a selection objection is the same measurement on rows the search never saw, so we ran one. The out-of-sample set is the PGL 6th-edition IDAC database scored by the same deployed closure on VT-2005 profiles (Table 2): another group’s compilation, sixteen times the broad set’s rows on a different profile database.

Table 3: **The map: every stratum the fixed estimator can bound**—the 15 of the 59 with $n \geq 40$, at the pre-declared cell (eight equal-count bins of $g(z^*)$, unbiased within-bin variance, row unit, residual-only convention), applied identically to all of them. *full* is the same cell under the full combinatorial convention. Every margin, and the MSE and bound behind it, is in squared $\ln \gamma_2^\infty$. The last column states what the row shows or the test it fails. The other 44 strata, the 24 of them carrying no estimate at all included, are in §S3.8 at this same cell.

Stratum	n	pairs	sources ^a	margin	90% margin ^b	full	cells ^c	what it shows, or the test it fails ^d
<i>Solvent class</i>								
glycol ether	182	43	3 (P8, 0.53)	+2.04	[+1.25, +2.87]	+2.23	4	the closure’s own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39] ^{0.7}	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194
aprotic acceptor	57	57	4 (P7, 0.32)	−0.19	[−0.4, +0.1]	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
<i>Solvent class, coarse</i>								
H-bond donor and acceptor (O)	328	75	8 (P8, 0.40)	+0.53	[−0.2, +2.2]	+1.34	3	the margin changes sign when one publication is deleted: P15 takes +0.527 → −0.417
H-bond acceptor only	70	59	5 (P13, 0.96)	+1.43	[−0.30, +6.21] ¹⁵	+0.77	0	the margin changes sign when one publication is deleted: P13 takes +1.434 → −0.192
<i>Solvent family, fine</i>								
glycol ether	182	43	3 (P8, 0.53)	+2.04	[+1.25, +2.87]	+2.23	4	the closure’s own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	[−0.15, +0.39] ^{0.7}	+0.57	1	the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194

^a Number of source publications, then the largest one and its share of that stratum’s squared error. The keys:

P1 10.1016/j.fluid.2013.06.028, P2 10.1016/j.fluid.2015.01.010, P3 10.1016/j.fluid.2015.11.037, P4 10.1016/j.fluid.2016.08.030, P5 10.1016/j.fluid.2016.10.009, P6 10.1016/j.jct.2013.05.006, P7 10.1016/j.jct.2015.12.034, P8 10.1016/j.jct.2016.10.013, P9 10.1016/j.jct.2018.05.003, P10 10.1016/j.jct.2019.05.011, P11 10.1016/j.tca.2012.02.005, P12 10.1016/j.tca.2015.11.010, P13 10.1021/acs.jced.7b00114, P14 10.1021/acs.jced.9b00726, P15 10.1021/je020196d. ^b Two-way solute × solvent cluster bootstrap, 5th and 95th percentiles. Row-identical strata are one row set:

drawn once and printed once, under the row set’s first name above and restated under the rest (10 of 43 row sets, 16 rows). Each prints at the precision its own draw count supports, decided cell by cell over 200 seeds and verified on the run that bought it. 2 carry two decimals (organic solute; glycol ether, the latter at 1.2×10^8 draws) and 9 carry one, both endpoints of each standing four Monte-Carlo standard errors clear of the nearest rounding boundary at that precision. For 11 no precision this table can print is determined: those carry two decimals with a superscript giving their endpoints’ Monte-Carlo standard deviation in units of that last digit, and the superscript is the warning rather than a refinement. A further 2 (aldehyde; halogenated | water as solute) no seed moves at all—each has one cluster on one of its two axes, so the resample is one-way over three clusters and about 10 distinct compositions exist: a bootstrap with nothing left to resample, not a converged one. The per-cell verdict, its draw count and its cost at either precision are on the deposited record (§S3.8). ^c In how many of the four unit × convention cells the stratum is both boundable and sign-stable under deletion of each contributing publication; the rule of §2.5 requires all four, which six of the 59 satisfy—three distinct row sets. ^d The condition the row is in, not a label for it. A one-source cell cannot be tested and a deletion leaving fewer than 16 rows leaves the cell undefined, so both read as untested rather than as passes. A margin is one-sided: positive puts $B_{\text{closure}} > B_{\text{insuff}}$ on those rows at that cell, non-positive is a failure to separate and licenses nothing. ^e The search behind every row, in the two nulls of the paragraph below. Under the binning bound of record, a chemistry-blind relabelling of the same molecules certifies at least one row set in 58% of 2000 draws, reaches +2.04 or more in 10.1% of all draws and in 17% of the draws that certify anything, and separates the six admissible strata from chance ($p = 0.01$) but not the two admissible positive row sets ($p = 0.286$). Under the pre-declared cross-fitted estimator, on the 473 rows it scores, the same relabelling reaches the observed maximum in 13.7% of all draws and in 54% of the certifying ones, against 9.4% and 16% for the binning bound on those same rows.

Table 4: **The map, solute axis**—the rest of the fifteen boundable strata, at the same estimator cell and under the same rule as Table 3, whose notes apply here unchanged. The alkane row is the second admissible positive row set, and clause (c) demotes it to the glycol ethers.

Stratum	<i>n</i> pairs sources ^a			margin 90% margin ^b	full cells ^c	what it shows, or the test it fails ^d
<i>Solute role</i> organic solute	440	179	14 (P8, 0.35)	+0.18 [−0.23, +1.35]	+0.82	2 the margin changes sign when one publication is deleted: P15 takes +0.184 → −0.094
<i>Solute family</i> alkane	129	45	7 (P15, 0.53)	+1.38 [+0.8, +2.3]	+1.72	4 the sign holds under every deletion, but 84% of these rows are the glycol ethers’ and the 21 outside give −0.22, so this margin does not stand outside the glycol ethers: demoted to them by clause (c); 10% shared with the aprotic acceptors
aromatic hydrocarbon	71	32	7 (P15, 0.53)	−0.19 [−1.0, +0.7]	−0.07	0 the margin changes sign when one publication is deleted: P6 takes −0.188 → +0.081
ketone	63	11	3 (P14, 1.00)	−0.53 [−0.85, +0.63] ^{1.8}	−0.33	0 too few rows remain after deleting P14 (1 of 3 deletions), so the sign is untested there
<i>Solvent class</i> × <i>solute role</i> glycol ether organic solute	182	43	3 (P8, 0.53)	+2.04 [+1.25, +2.87]	+2.23	4 the closure’s own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water organic solute	111	15	3 (P14, 0.75)	+0.05 [−0.15, +0.39] ^{0.7}	+0.57	1 the margin changes sign when one publication is deleted: P2 takes +0.046 → −0.075; P14 takes +0.046 → −0.194
aprotic acceptor organic solute	57	57	4 (P7, 0.32)	−0.19 [−0.4, +0.1]	−0.15	4 the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
<i>The set as a whole</i> whole set	477	185	15 (P13, 0.32)	+0.51 [−0.1, +2.2]	+0.95	3 passes here and fails at the pair unit (residual-only convention), where deleting P13 takes +0.270 → −0.119

The row set, the estimator cell, the admissibility rule and what would count as confirmation in each direction were written down before any margin was computed (§S3). **The chemistry could not be tested there**, for a reason about the data rather than about the number: the cell the finding is about—a glycol-ether solvent with an *organic* solute—has no rows there. *Not testable* is the third verdict the pre-declaration lists and counts as non-confirmation, and it follows from the row inventory alone: what the pre-registration delivers here is the absence of confirming data and not a failed replication. The nearest cell that database does carry is water in glycol-ether and polyol solvents, and it is inadmissible under the same rule that governs the map—the rule refuses it out of sample for the reason it refuses water-as-solute in sample—so its margin licenses nothing and the sign is never put at risk. Over that database’s ten solvent classes, stated as description and not as a test (Table S10), no class both passes the rule

and separates the two terms, and every margin above the one admitted positive class sits on a class the rule refuses. Out of sample the glycol ethers are neither where this closure fails hardest nor a cell the rule can read. The chemical localisation is therefore what the instrument licenses on the rows it was computed on, and the title does not carry it.

It is also the whole of the separation this paper reports. The phrase “COSMO-SAC error exceeds input error” would carry it past five boundaries, and each of them is a measurement here. No other stratum of the map returns a separation (below). Neither data set taken whole does: the aggregate is retired above, and the VT-2005-matched set’s +0.05 does not stand (§3.5.2). Nothing on the solubility axis does, where the two terms do not separate at all and where the substitution of §3.1 was measured. The error that stands above the bound is COSMO-SAC’s *as deployed*, the composite of map and computed conformer that the pipeline

runs on rather than the kernel (§3.5.3). And the margin is one its own chemistry-blind null returns at the ninetieth percentile, on the one chemistry no out-of-sample rows exist for.

Which chemistry does not separate. Three of the six admissible cells are that same row set at three granularities. The remaining three are two row sets and neither establishes anything: the alkane solutes are admissible and positive and clause (c) demotes them to the glycol ethers, which hold more rows and hold most of theirs, while the acceptor-only aprotics pass the rule and license nothing, the instrument being one-sided. That is the whole of the uniqueness claimed for the glycol ethers: the only row set admissible, positive and left standing by clause (c)—*not* the only admissible stratum and not the only one sign-stable under every deletion. Which cell each of the remaining strata fails on, and by how much—too few rows to bound, one publication so that the deletion cannot be run, or a sign lost under a deletion—is in §S3.8, and none may be used to explain a reversal. The whole-set row passes at the fixed estimator cell and fails once the pair unit is required, so the rule does not by itself dispose of the aggregate there; the composition argument above does.

Why the output is a map. $B_{\text{insuff}}^{\text{up}}$ is nearly stratum-independent and MSE is not. Across the three classes the design can bound—glycol ether, water and the acceptor-only aprotics—the bound spans a factor of 2.0 against errors spanning a factor of 25. Adding the three that carry an estimate without being boundable gives 6.8 against 51; that 6.8 is set by a mono-alcohol figure the artifact flags as an invalid bound on a stratum already excluded. The two orderings are unrelated, the glycol ethers carrying the lowest bound and the second-highest error among those six. Against 2000 random partitions of the broad set’s 477 rows into blocks of the observed sizes of its nine solvent classes, at the same cell, the chemical partition’s range (largest minus smallest) in MSE exceeds every draw while its range in the bound is unremarkable ($p = 0.26$); their ratio gives $p = 0.02$. Every stratum’s figures are in §S3.8. That is the estimator showing through: every stratum is

summarised by eight quantile bins of one scalar, so the bound measures the residual spread of m inside such a bin, a property of the binning rather than of the chemistry, while nothing constrains the error to follow it. An aggregate margin is therefore a composition-weighted average of a term free to move 25-fold offset against one that barely moves. It will report a binding closure for any set whose composition happens to include a chemistry the closure fails on—a statement about the aggregate, not about the bound, which stays valid in each stratum where it is computed.

3.5.2 The VT-2005-matched set ($n=60$)

This set carries the paper’s other reference-matched analyses, and it is far weaker. All 60 rows sit at 298 K, so z^* is the profile pair alone and $\dim(z^*)/n \approx 102/60$ leaves the split behind one-sided bounds; it is strongly unbalanced, methanol occupying 25 of the 60 rows; and its 17 solvents fall in six of the nine classes, none reaching the $n=40$ a bound needs, so it supports no stratified bound and no stratum of it is stated as a finding. Its aggregate margin is $+0.05$, and it does not stand: it fails under deletion of the two leverage pairs, under dropping methanol, under blocked out-of-fold estimators, and under 11 of the 60 single-pair deletions, where on the broad set no single-pair deletion moves the sign at all (§S3). Zero-mean label noise can understate a separation but not create it (§2.1.1); what a margin here would bound is a label systematic tracking z^* , and that is *not excluded* under the unbiased within-bin variance (§S3). Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are not independent samples—this is the sub-design VT-2005, and therefore the paradox’s own oracle, is defined on, and the agreement between the two sets’ error orderings is a consistency check rather than corroboration.

3.5.3 Whether a better closure moves the ceiling, and where this one fails

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 $\rightarrow$ 2010 step delivers one on the VT-2005-matched set and not on the broad IDAC set, and a block split identifies the factor that carries that reversal. On the smaller set the step more than halves the error under the full convention ($P > 0.95$), the convention under which the two kernels’ published error figures are reported, and cuts it 42% under the residual-only one. On the broad IDAC set (477) none of its three cells establishes an improvement, and none measures an equality either. The step scores -0.99 at $P \approx 0.2$ under the residual-only convention and -0.07 at $P \approx 0.6$ under the full one; with the dispersion term on it scores $+0.06$ at $P \approx 0.6$ under the full convention, a point estimate on the improving side and no better resolved than the two that are not. The broad set leaves the sign open in either direction and does not measure it at zero.

The reversal is the *chemistry* and neither the profile database nor the temperature range. The block split that says so runs with UD profiles, one extraction and one convention throughout. The 2010, *dispersion off* arm wins on the 57 pairs shared with the smaller set ($1.78 \rightarrow 0.77$ at $P > 0.95$, full convention); it does not win on the complementary 420; and it still does not win on the 80 complementary rows that are *also* at 298 K. That third block is the one the temperature half of the claim rests on and the weakest of the three, at $P \approx 0.3$ and with no interval, and the three blocks were never tested against one another. The 57 shared pairs are broad-set rows as well and therefore not an independent sample of that set. That block split is the whole of the evidence for the reversal being chemical. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set’s chemistry**, and that null is additionally a small-molecule and glycol-heavy one, so on drug-like solids the lever is untested rather than absent.

Under the 2010 kernel’s own typed latent,

306-dimensional and so several times worse resolved, the floor checks return two aggregates of opposite sign. They are $+1.29$ on the broad IDAC set ($P > 0.95$, ≈ 0.9 , ≈ 0.9 over pair, two-way and source clusters) against -0.77 at $P \approx 0.3$ on the VT-2005-matched set, and only the first of them stands. That is loss of resolution and not a demonstrated flip to the inputs, a flip needing a *lower* bound on B_{insuff} . The input-fixed control and its three limits, the three arms on both sets under both conventions (Table S8), and the third block at 298 K are in §S6.6.

Where B_{closure} dominates—on this set, the one row set the rule leaves standing—the ceiling is in the map *as it is deployed*: the fixed kernel together with the one computed conformer it reads, which is the composite B_{closure} is the error of (§2.1.1) and the object the pipeline runs on. Which of the two carries the error is a question this section does not answer; nor does dominance say the composite is *fixably* wrong. A small structure-preserving residual was fitted on the fixed COSMO-SAC exchange-energy kernel and five-fold cross-validated on the 60 pairs of the VT-2005-matched set, on their reference VT-2005 profiles. It removes $46 \pm 5\%$ of the reference-input error under random folds but $-5 \pm 16\%$ once folds are grouped so that no solvent is shared. Only the grouped design is free of leakage, the closure error being predominantly a between-solvent systematic. It gives a null and not a bound: the $\pm$ prices only optimiser noise over five fit seeds, the held-out fold spread that dominates it going unpropagated (§S3). B_{closure} is therefore an error a handful of kernel parameters can absorb *within* a solvent and that we have not shown to be correctable across solvents. Where it fails is chemically placed, and placed differently on the two sets. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents and the closure is near-exact on acceptor-only ones—the chemistry COSMO-SAC’s documented single-parameter H-bond term governs, the term the donor/acceptor-typed 2010 closure was built to replace.^{27,32} That the 2002 kernel fails on hydroxylic and aqueous systems is es-

tablished, and the localisation restates it rather than discovering it; what is new is that on one such chemistry the failure is bounded against what the closure’s inputs cannot resolve. Half of that reading carries to the published evaluation of Table 2 and half does not, and on the broad set neither it nor its replacement reproduces, so we state neither (§S6.6). It places the error in a *chemistry* and not in one component of the composite: the reference profile for each of those solvents is one gas-phase conformer too, and neither set separates a displaced reference from the kernel. One part of that is now bounded rather than open. Replacing the four glycol-ether solvent references with Boltzmann conformer ensembles computed at BP86/def2-TZVP with CPCM moves the stratum’s margin from +2.04 to +2.27, 90% interval [+1.36, +3.29], and an equal-norm random displacement leaves +1.68—so the separation is not an artefact of those references’ conformations, though displacing the solutes by the same norm moves it as much. Reference uncertainty at large is not bounded by that, and its scale is measured: two defensible calculations of the same rigid molecule’s profile differ by a median relative- L_2 of 0.282, four to eight times the conformational displacement, and substituting a recomputed lowest conformer for the tabulated one moves this stratum’s margin by +5.43 against the ensemble’s +0.23. Every substitution reported in this paper is therefore scoped to the VT-2005 tabulation’s level of theory (§S6.6).

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualification of the result: it is what a practitioner adopts by default, a closure’s fidelity is *not knowable in advance* on a new system, and the field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. What follows when it is not is that the learned latent silently departs from the quantity it is named after (§3.3), invisibly without an external oracle. Outside that one row set the map reports cells it cannot resolve, not cells in which the closure is sound. One arithmetic sizes that exposure, and it is worth stating because the ground truth

charging B_{closure} is quantum chemistry and not experiment. One closure read on two profile databases—HF-TZVP against BP-TZVPD-FINE—differs by a factor of 2.6 in AAD (Table 2). The glycol-ether stratum it is set against is computed on the UD profiles the broad set is matched to, and its margin closes only under a more than ten-fold reduction in its squared error; a profile source better than UD by the whole of that published bracket would leave that sign standing. The two sides therefore name different databases, and the bracket enters as the size of a database-to-database effect rather than as a comparison either side is on. The arithmetic holds the insufficiency bound fixed, which a new profile source would not, so it sizes the exposure and does not bound the share; the working, and the route by which a displaced reference loads onto B_{closure} through the curvature of g even where g is exactly specified, are in §S6.6 and §S8.

3.6 The same split on a second closure: pKa through a fixed Hammett relation

The instrument is not specific to COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship $g(z^*) = \text{p}K_{a,0} - \rho \sigma_{\text{H}}^*$ in a substituent constant, with $\text{p}K_{a,0}$ and ρ the tabulated Hansch–Leo–Taft reaction-series constants, one literature pair per scaffold—(4.20, 1.00) benzoic acid, (9.99, 2.23) phenol, (4.60, 2.77) anilinium, (5.25, 5.90) pyridinium—never fitted or refit on these data, so g is fixed in the sense Lemma 1 requires and the same g runs inside every split. It reads two things, so z^* is the pair (s, σ_{H}^*) : the series s , which selects that literature pair, and the substituent constant at the tabulated value the rule below assigns. Only the constant is learned—the learned intermediate $\hat{\sigma}_{\text{H}}$ is that constant predicted from structure; the external reference is the Hansch–Leo table,³³ printed in full in §S7; and the data are the OPERA experimental compilation³⁴ under a uniform, closure-independent quality

filter that drops pre-ionized microstates.

The reference constant is assigned mechanically, by a rule that reads structure alone and never the residual (§S7), and that rule is also what defines the two poles. A record enters under one of the four reaction series—benzoic acid, phenol, anilinium, pyridinium—which fixes its $(pK_{a,0}, \rho)$; σ_H^* is then the sum, over the substituted ring positions, of the Hansch–Leo constant of that substituent *at that position*, σ_m meta and σ_p para, taken from a fixed list of nineteen. An ortho position carries no tabulated constant and enters the sum as zero, and so does a substituent outside the list. The first pole—the *clean* one below—is the 158 records whose every substituted position is meta or para with a tabulated substituent; the *degraded* pole is the 846 where at least one position is neither—571 of them carrying an ortho substituent, the other 275 an untubulated one—and there σ_H^* is the partial sum over the positions it can see. The split is not benzenoid against heteroaromatic: all four series occur on both sides, 32 of the clean pole’s records being pyridinium.

The two classic failure modes of that relation fall onto the two channels of Lemma 1: *resonance saturation*, a curvature the linear closure cannot represent, lands in B_{closure} , while what the zeroed positions carry—ortho field and steric effects, and whatever an untubulated substituent contributes—is information the intermediate lacks and lands in B_{insuff} . The two poles differ accordingly. On the clean one, the chemistry the relation was built for ($n=158$), the closure predicts to RMSE 0.55, and the two available bounds on B_{closure} disagree about what that leaves. The binning-free floor $\text{MSE} - B_{\text{insuff}}^{\text{up}}$ is -0.2 (95% CI $[-0.4, -0.0]$), and it establishes nothing about this closure: the out-of-fold $B_{\text{insuff}}^{\text{up}}$ is 0.500 against an MSE of 0.302, so that difference is negative whatever the closure, and what those 158 rows across four series show is that a flexible regressor on the closure’s own argument predicts worse than the closure does. The Lemma 1(b) bound is informative and runs the other way. The per-scaffold systematic offsets reach 0.429 in pK_a on phenol (+0.303 anilinium, +0.034 benzoic acid, -0.198

pyridinium), and the n -weighted mean of their squares bounds B_{closure} below at 0.083, 27% of the 0.303 total, and at 0.184 on phenol alone, half of that series’ own 0.371. At least a quarter of the clean pole’s error is closure error, so the pole is not shown to be well specified. The quality filter moves the floor and not that conclusion: the rule is closure-independent—net formal charge $\neq 0$, applied uniformly to both poles, never reading a residual—but it drops five pre-ionized microstates, and on the 163 rows before it the same floor is $+0.55$ rather than -0.2 , positive and establishing $B_{\text{closure}} > 0$ by the other route as well (§S7). On the degraded pole ($n=846$) it degrades to RMSE 2.26, of which the out-of-fold within-series regressor places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure—both figures one-sided, in the same direction as the solubility bounds. The filter leaves that pole where it was and moves the other by a factor of eleven in MSE, 3.33 to 0.30: before it the two poles differ by half an order of magnitude and after it by more than one, so a large part of the contrast between them is what the filter makes. The reversal itself does not rest on that—it is a sign, measured within each pole—but the poles’ separation in level does, and so does the sign of the clean pole’s floor.

That estimator is not the eight-bin estimator cell of §2.5, and the functional is not the same either: the two figures just given are floors $\text{MSE} - B_{\text{insuff}}^{\text{up}}$ on B_{closure} , which when positive establish $B_{\text{closure}} > 0$, and not the separation margin $\text{MSE} - 2B_{\text{insuff}}^{\text{up}}$ of Table 1, which when positive would establish the stronger $B_{\text{closure}} > B_{\text{insuff}}$. Neither coordinate of $z^* = (s, \sigma_H^*)$ needs binning— s takes four values and σ_H^* is a scalar—so the conditional variance is estimated directly, as the *out-of-fold* residual mean square of a random forest of pK_a on σ_H^* fitted *within* each series, five-fold and pooled over the series by n , which exceeds B_{insuff} by the regressor’s own approximation error and so leaves a lower figure on B_{closure} . Conditioning on s is not a free choice and it is not the conservative one: pool the four series and the same estimator returns 8.3 against the degraded pole’s MSE of 5.1, a conditioning under which g is not measurable,

so Lemma 1 does not license it and the closure floor it leaves would in any case be vacuous. The floor of 0.6 therefore rests on s belonging in z^* , which is what the lemma demands of a closure that reads it. Those folds carry no leakage of the kind §3.5.1 refuses a fitted estimator for on solubility, where a held-out row borrows strength from a near-twin through the representation: here the regressor’s only input is σ_H^* within a series, so no fold can hand it what the closure’s own argument does not already carry, and the identity asks of the fold design only that a row be held out of the fit that predicts it.

The decisive test is the same oracle swap, measured *in-distribution* across $K=20$ within-scaffold splits: the learned $\hat{\sigma}_H$ replaced by the tabulated σ_H^* in the same closure—the series label is read from the structure in both arms, so the swap moves one coordinate of z^* —the sign of the change recorded per stratum. It flips. On the clean pole ($n=158$) the reference *improves* accuracy—MAE 0.48 learned against 0.315 with the reference, a gap of -0.165 at a 90% CI of $[-0.28, -0.04]$, better in 19 of 20 splits and in three of the four reaction series—so the ceiling there is the noisy learned estimate and not the closure, which is an internal positive control on real data. It is a control on the swap’s sign and not on the closure’s specification: the Jensen bound puts at least a quarter of that pole’s error in the closure, so the reference wins there against a closure carrying error of its own. On the degraded pole ($n=846$) the same swap *degrades* it—0.83 against 1.62, worse in 20 of 20 splits and in all four series, a gap of $+0.79$ at a 90% CI of $[+0.57, +0.95]$ —the freely learned constant absorbing what the zeroed positions carry and the tabulated one lacks. The sign of grounding therefore reverses inside one closure, in the direction the two poles’ construction fixes in advance—which ring positions the Hansch–Leo table can see, complete on one pole and a partial sum on the other—with Remark 1 supplying the vocabulary and not the prediction. That the reversal runs through the insufficiency channel rather than through the closure-error cancellation of §3.3 is unsupported on the clean side, where the Jensen bound above leaves

at least 27% of the error in the closure. On the degraded side the attribution rests on the flip itself—the freely learned constant beats the tabulated one there, 0.83 against 1.62, in 20 of 20 splits—and on how that pole is built, σ_H^* being a partial sum over the positions the table cannot see. What the Jensen bound adds there is that it does not refute that reading: it leaves at most 87% of the error in insufficiency, and an upper bound is as consistent with none of it as with all of it. The out-of-fold regressor two paragraphs above is a different estimator answering the same question on the same pole, at 4.5 of the 5.1—and it is an upper bound too, so the two agree without either of them supporting the attribution. Both signs are one comparison: the sign of the grounding gap is by construction that of the learned arm’s error against the fidelity-set oracle error, and what a competence sweep adds is that the oracle side of it does not move with the training budget while the learned side does. The clean pole’s threshold (≈ 0.315) is never beaten by the learned arm, so grounding helps at all four budgets swept, while the degraded pole’s threshold (1.62) is high enough that the learned arm beats it across the whole interpolation range, down to 12% of the data, and grounding reverts to helping only under scaffold *extrapolation*, where the learned constant breaks (2.6 ± 2.0) back above the threshold—in the mean over the six splits, three of which still have grounding hurting. **The flip is an in-distribution result and the scaffold protocol does not certify it.** Re-running the same $K=20$ certification on a scaffold split rather than a stratified one, the reference helps on *both* poles: the clean pole’s gap is -0.61 and the degraded pole’s -1.10 , with a series-cluster interval of $[-1.86, +0.37]$ and $P = 0.28$ there, so the degraded pole’s “grounding hurts” half does not certify and the reversal does not either. Every solubility result in this paper is measured on a scaffold split, and this is the one place the protocol changes; what the scaffold run confirms is the mechanism rather than the flip, the learned constant breaking under extrapolation and rising above the oracle threshold on both poles. The clustering unit is the

Hammett reaction series, of which only $G=4$ exist, so the bootstrap resolves nothing finer than series-level sign agreement, read as a sign test on four units: $p=1/16$ on the degraded pole, where grounding hurts and all four series agree, and $p \approx 0.31$ on the clean one, where three of them do—so that test carries the degraded pole and not the positive control, which rests on its splits and its gap instead. The construction, the estimator validation and the measured fidelity sweep are in §S7.

3.7 Scope, accuracy, and limitations

The result is a cautionary measurement about one of the two operations the word “grounding” names, and it is two measurements made in different places. On solubility: supervising a learned physical latent on reference values during training helps, and *supplying* those values at inference hurts. On infinite-dilution activity data, and there on one class of solvent: the fixed map’s own error stands above the bound on what its inputs fail to carry—the condition under which better inputs surface a map’s bias rather than help, and one the standard route to a better closure did not clear either. The two are measured in different regimes on largely disjoint chemistry, no measurement here connects them, and what would settle it is named in limitation (i) below and in (x); until it is run, “the substitution hurts *because* the map is misspecified” remains untested. Both sets are scoped by the reference database that defines them and neither is the deployment regime. The useful form of the finding is one-sided and narrow: a place where grounding effort is *not* the remedy, on one row set of the map, rather than an allocation across chemistries.

The unconstrained control’s representation. If the physics arm’s intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is not. A ridge probe of the control’s pair representation recovers the model’s own prediction at $R^2 = 0.98$ but not

the measured ideal-solubility term. It is scored on 13 held-out solutes / 527 rows at zero solute overlap, and that term is 92.2% between-solute variance, so the effective n is 13 clusters and not 527 rows. The three seeds give $R^2 = -0.506/+0.111/+0.027$ at 42/43/44, with permutation $p = 0.894/0.055/0.177$ against a 1000-draw null. Their between-seed range of 0.616 (sd 0.334) is larger than any one of the three values, so a single seed would have been unreadable and seed 43 alone would have read as a near-miss. What the negative supports is therefore not an exclusion, and the level below which it excludes nothing is not computed here. What the deposit does fix is the null’s shape: its 95th percentile is 0.105/0.117/0.096 over 1000 draws and its maximum 0.316/0.380/0.346, and each of the three measured values sits at or below its own null’s ceiling—which is what the three permutation p above say in another currency. The design therefore excludes nothing below that ceiling, and no power calculation was run to say how far above it the design reaches. No comparison is drawn between the between-seed spread and the null’s support: the null is two-sided and reaches below the lowest measured value, so its support is the wider of the two. That is limitation (ix), and two things fix its size rather than the encoder’s. The 13 solutes cannot be enlarged, the reference scarcity of §2.2 being what caps them, and the temperature probe rests on the same 13. And the crystal-slope test is capped by the corpus rather than by the probe: this corpus’s empirical $\ln x_2$ -against- $1/T$ slope carries $-\Delta H_{\text{fus}}/R$ at $r = +0.24$, so the quantity the probe is asked to find is itself weakly present in the labels it is scored against. What the representation is organised by instead is scaffold identity, whose share of the representation’s variance sits at the solute-identity ceiling—on a split where those two partitions nearly coincide by construction, only 11 of the 124 test scaffolds holding more than one solute, so they agree on 91% of the data and a share at that ceiling is weak evidence of within-scaffold indistinguishability rather than a measurement of it. A test molecule on an unseen scaffold is not a small extrapolation but a point off the axis the rep-

resentation is organised along. That is a description of this representation and not a diagnosis of the encoder, which the linear probe of §S6.1 does not supply. The full tables, the controls, the direct within-scaffold measure on the solutes that do share one, and the temperature probe are in §S9.

The contribution. The unit of contribution is the measurement, not any single number it returned. Its two ingredients are individually standard. The error a composed predictor makes when it is handed reference inputs divides exactly into two parts, what the fixed physical model gets wrong on those inputs and what the inputs themselves leave undetermined about the measured quantity; that division is the law of total variance applied to a composition. Whether replacing a learned input with an accurate one helps or hurts follows from which of the two parts is the larger, which is the model-discrepancy principle of Kennedy and O’Hagan.^{20,21} Combining them gives an instrument. It divides the error of a physical model that is never refitted to the data, bounds the model’s own part from below and the inputs’ part from above, and predicts the sign of grounding’s effect before the substitution is made, in both directions. Acid dissociation supplies that prediction as a sign that reverses inside a single data set, in distribution: the tabulated constant helps on the molecules its table describes completely and hurts on the rest. A learned concept in a concept-bottleneck model admits no such test, because nothing tabulated exists to put in its place.^{18,19} The solubility results are the instrument’s first application. Both scopes are narrow: the sign reversal is certified on a stratified split and does not certify on a scaffold split, which is the protocol every solubility result here is measured on, and it resolves no finer than the four reaction series it is read on; the solubility results’ transfer out of the matched regime is not established.

Where the accuracy stands, and the confound in it. Table 5 places every arm of this work beside the external models, in two blocks that may not be read across. The

physics arms trail their own control in the seed mean—the COSMO-SAC arms at every seed, the NRTL arm at two of three, being ahead at seed 42 (1.734 against 1.749; Table S3)—and the predict-the-mean row shows how little room sits above them. The physics and control arms were tuned separately and no hyperparameter-matched control was run, so the global Δ MAE of +0.14 between them on the 5608 labelled test rows—from the unrounded means, where Table 5’s rounded cells differ by 0.15—carries no information about the thermodynamic bottleneck, and neither do the per-solvent-class map, the per-solute-class map, the novelty inversion or the data-efficiency trend, all of which resolve the *same* difference along chemical axes. We claim no accuracy result in either direction; those strata are reported in §S6.4 and §S6.5 as descriptions of where two particular pipelines differ, and no conclusion rests on them. Running the matched control is the direct repair and we have not run it. What survives the confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning: §3.1, §3.2 (the ranking result included), §3.3 with the output correction folded into it, and §3.5, which uses no trained model at all. Separately, and not as an accuracy claim about the bottleneck: no configuration we ran attains a lower *seed-mean* error than the control—at one seed the NRTL arm does—and our one attempt to move the ceiling by grounding the crystal factor on external labels is underpowered rather than null. 97.76% of its pool was melting points the model was already trained on. Where the pool did carry something new the crystal head moved: its paired T_m MAE over the pool itself went -12.233 K $[-18.25, -6.27]$ on the 345 net-new molecules, against -0.371 K $[-0.81, +0.05]$ on the 15,082 duplicated ones. On the held-out solutes the same head is the worse of the two: T_m MAE $45.0 \rightarrow 46.6$ K over the 2518 test solutes carrying a label, a paired $+1.59$ K with a solute-clustered 95% interval of $[+0.31, +2.85]$. The downstream solubility half is weaker still than its row count suggests: the $+0.065 \ln x_2$ MAE gap against the grounded arm is 5608 rows from 147 test solutes, and clustered by solute it is

+0.088 with a 95% interval of $[-0.127, +0.284]$, crossing zero (§S6.3). The one place the physical *structure* helps is temperature extrapolation, and it helps there as a property of the hypothesis class rather than of any arm trained here. Fitting below $T \leq 309.75$ K and scoring at $T \geq 330$ K—the same 1751 solute–solvent pairs on both sides, so the split withholds temperature and not structure—a structured van’t Hoff class, log-solubility affine in $1/T$, reaches an MAE of 0.37 over the 3343 high-temperature test rows against 1.29 for a random forest over Morgan fingerprints and temperature, both in $\ln x_2$ units. Neither figure carries an interval: each is one fit on one fixed temperature cut, with no replicate behind it. Whether a trained arm inherits that form we do not know: the only checkpoint ever scored on the temperature-withheld split is one whose predictions are near-constant, so it is untested rather than answered (§S6.1, §S5).

The external block is not like-for-like and the table says so in its own block heads. FastSolv and SolProp are deposited here on a *by-solute* split, this work’s arms on the harder solute-scaffold split, and the published figures are as published, on other corpora and in another unit; scoring released FastSolv weights on our test rows is invalid because FastSolv is trained on this corpus. So there is no like-for-like external comparison in this paper. That is a budget and not a barrier: the FastSolv harness deposited here retrains from a train/validation/test triple and accepts this split’s own files, and the run was not made, so the one number that would place these arms against published work on the split they are argued on does not exist. What remains is a scale and not a placement: this work’s control reaches $\log_{10} S$ RMSE 1.00 on held-out scaffolds and its physics arms 1.39 to 1.60, against published figures of 0.83 to 0.99 for that evaluation’s leaders and 1.43 to 2.16 for the physics-informed cycle. Those are on other corpora and on splits that are not this one, and no row is shared with them, so no ordering follows from setting them beside each other. Those six published figures are $\log_{10} S$ RMSE in mol L^{-1} from a single evaluation, Attia et al.,¹ on two

corpora: the SolProp test set, whose solutes are disjoint from its training corpus, and the near-ambient Leeds set of Boobier et al.³⁵ The leaders are that work’s two direct models—the solution-FASTPROP ensemble at 0.83 and 0.95 and its solution-CHEMPROP companion at 0.83 and 0.99—and the physics-informed cycle is SolProp,² scored in the same evaluation at 1.43 and 2.16. Each range is two point values on two corpora rather than a spread, and none of the six carries a published uncertainty. What may be read from those figures is what survives the clip sweep and not the two decimals; the sweep, the random-pair rows and the conversion behind the $\log_{10} S$ column are in §S4.

The sign rule’s own prediction. Remark 1 predicts *when* to expect these negatives: a prior hurts precisely in the mature-data, low-noise, well-covered regime. The synthetic sweep (Fig. S9) is true by construction there, so it verifies the instrument rather than the hypothesis, while the pKa closure exercises the sufficiency channel on real data. Whether grounding helps is shaped by *two* axes, closure fidelity and how the latent is trained; that plane is not characterised on real systems here. The chemical strata form a pattern in the direction the sign rule would give, but separately tuned pipelines can produce it and there is no multiplicity control, so §S6.4 records it as the experiment a matched control should run and the diagnostic is *proposed* rather than validated.

Limitations: what the sets can carry.

(i) The phenomenon and its causal attribution lie in *different regimes*: the paradox is measured where the model operates (finite composition, all γ , $n=5608$), the attribution and the latent-departure result only on infinite-dilution activity data ($n=477$ across temperatures, and the 298 K, low- γ , VT-2005-matched $n=60$ and $n=44$ sets). Transfer back to the paradox’s own regime is a conjecture and is the principal open question here. The $n=60$ set is narrowly sourced—all by gas chromatography and all from three publications, 33 pairs from one—so neither a method-stratified nor a source-clustered re-estimate is available on it, and its

Table 5: Solubility accuracy in two blocks that may not be read across: this work on the solute-scaffold split, external models re-run here on an easier by-solute split. Two sample sizes appear where a row’s two metric columns are not scored on the same rows, one where they coincide.

Model	n ($\ln x_2 / \log_{10} S$)	MAE $\ln x_2$	RMSE $\log_{10} S$	R^2 $\ln x_2$
<i>This work, solute-scaffold split; mean $\pm$ s.d. over seeds, three unless the row says otherwise</i>				
DirectGNN, same encoder, no closure (<i>control</i>)	5608 / 5440	1.70 ± 0.03	1.00 ± 0.02	0.42 ± 0.03
TGNN, NRTL closure (physics)	5608 / 5440	1.79 ± 0.07	1.54 ± 0.07	0.34 ± 0.03
TGNN, COSMO-SAC closure, supervised $\hat{\sigma}$ (physics)	5608 / 5440	1.85 ± 0.05	1.39 ± 0.19	0.33 ± 0.05
TGNN, COSMO-SAC, reference σ^* substituted at evaluation	5608 / 5440	2.25 ± 0.02	1.60 ± 0.14	-0.03 ± 0.04
<i>Scale reference on the same solute-scaffold rows</i>				
predict the test-set mean of $\ln x_2$	5608 / 5440	2.32	1.33	0.00
<i>External models re-run on this corpus on a by-solute split, one run each—not a ranking against the block above</i>				
FastSolv, retrained on this corpus, re-scored	10287	1.44	0.84	0.55
SolProp cycle, released model + 3-parameter recalibration	10287	2.34	1.34	-0.18
SolProp cycle, released model, zero-shot	10287	2.36	1.87	-0.47
SolProp encoder retrained on $\ln x_2$, cycle discarded	10570 / 10287	1.62	1.07	0.39

margin is extraction-sensitive: a broader ThermoML pull at 298 K widens it but *inverts* it on the gas-chromatography subset, a sensitivity we leave unresolved. The broad set does not move that way under the same cut and is not the set the paradox’s own oracle is defined on; the arithmetic and the enlarged-pull values are in §S8.

Limitations: what the estimator can deliver. (iv) B_{insuff} is upper-bounded, not point-identified (no replicates at fixed z^*); the smoothness-free Jensen bound assumes m and g share the $\ln \gamma_2^\infty$ reference state, established by the closure validation of §2.3, so the mean offset is decoder bias and not a units artifact. B_{closure} is the error of a composite—the fixed map together with the calculation that produced z^* —and no measurement here divides it between the two; §3.5.3 sizes that exposure and leaves the share unbounded. (vi) The stratification reads structure alone, so no cell was selected on its outcome, and it is exhaustive and disjoint *within* an axis but not across the two axes or their granularities: 362 of the 1711 stratum pairs share rows, all 37 water-as-solute rows lie inside three solvent classes, and 108 of the 129 alkane-solute rows are glycol-ether rows, so cells from different axes are not independent evidence and those rows are counted once. Granularity is an analyst choice, which

is why three solvent levels and two solute levels are all reported. There is no multiplicity control on the per-cell bootstrap frequencies, so one cell’s P describes that cell alone, and the admissibility rule is a stability filter and not a correction for the search; six of the nine solvent classes are not boundable here and three carry no estimate at all. What the search survives is measured rather than assumed, and neither test establishes the finding as a chemical effect: under the chemistry-blind relabelling null of §3.5.1 the shape of the finding actually claimed—the number of admissible *positive row sets*, and the certified magnitude—is not above chance, and the pre-declared out-of-sample test of the chemistry returned *not testable*. The single finding is therefore stated at the level the instrument licenses on its own rows and *not* as a multiplicity-controlled chemical effect, and the same holds a fortiori for every cell the map reports and does not state. The rule is also one-sided in what it can deliver: with B_{insuff} bounded only from above, no cell can ever return “better inputs would help here”, so the map only ever withholds a chemistry from an allocation of grounding effort.

Limitations: what is untested, and what is not auditable. (xii) The scaffold-leak guard is a property of the released stream builder and not a certified property of these

runs, so the 0.20 supervision gain is the one reported number a leak could inflate and is un-certified until the leak-free re-run (§2.2); deleting the eight reachable solutes leaves $+0.17$ $[+0.03, +0.31]$, which bounds where a leak can act and does not restore auditability, and neither do the arguments that bound the other contrasts. (xiv) Two terms the thermodynamics asks for are set to zero in the deployed model, and both are absorbed at training time by the same learned σ head. ΔC_p is zero in the ideal term, one-signed and worth a median 0.32 in $\ln x_2$ where a melting point is measured (§2.1). The Staverman–Guggenheim combinatorial term is off in every arm the paradox is reported on, and the σ head emits a profile shape and a cavity area but no cavity volume, so evaluating that term needs a volume from elsewhere, as the one arm that restores it takes one. Both cancel in the evaluation-time substitution and neither cancels in the training that produced the drift of §3.3, so the mechanism there is not isolated to the activity kernel. The nine remaining limitations are in §S6.7, keeping their numerals. Every one of them is also stated in this document, and this index gives the place rather than the kind. Five are stated where they bite: (ii) what the convention rule costs the aggregate and (v) the constant-offset bound’s convention-specificity in §2.5, (iii) that the glycol ethers are not the only cell of the map to keep their sign in §3.5.1, (vii) the accuracy ceiling standing a factor of five above the label-noise floor in §2.2, and (xi) the separate-tuning confound above. Four print their content in the section that carries the result: (viii) what the three-seed latent-departure result establishes and what it does not, in §3.3; (ix) what bounds the black-box probe, in the probe paragraph above; (x) what the closure-fidelity contrast does not measure and what would settle it, in §3.4; and (xiii) what the deposit does not make checkable, in the data-availability statement. What §S6.7 adds to all nine is the detail each is qualified by—which two cells, which magnitudes, which of the probe’s tests—and none of the nine is there alone.

4 Conclusions

Grounding a learned physical intermediate in reference values is two operations, not one, and here they had opposite signs. The operation that failed is the one a practitioner reaches for, because it is the one that needs no re-training. The intermediate paid for the arrangement. Trained through a model it cannot correct, and measured on runs of its own, it stopped reporting the physical quantity it is named after wherever a reference existed to hold it against—which is the price of reading a grounded latent as a measurement.

The general statement is a design rule with a test attached. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures whether the fixed model itself limits accuracy, and where it answers, the answer decides the remedy: a more faithful physical model—the calculation that supplies its reference values included. The answer comes chemistry by chemistry rather than once for a field, and it comes one-sided, so what the measurement returns is the chemistry on which better inputs are not the remedy. What this work establishes about the instrument is the map itself: which cells it can answer in, and at what size. Whether a latent trained through an imperfect model becomes, in general, a surrogate for that model’s error is the prediction this work leaves testable.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data and Software Availability

The analysis code, and those intermediate artifacts small enough to version, are available at <https://github.com/doctawho42/tgmn-solv> (commit [hash to be supplied at submission]) under the MIT licence, that commit being the one the reported numbers were produced from; the model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts will be deposited in a Zenodo archive, [whose DOI is registered at submission]. The data sources are public: BigSolDB 2.0²⁴ for solubility, the VT-2005 database for the reference σ -profiles, Ref.³² as redistributed with Ref.²⁶ for the UD profiles, ThermoML for the infinite-dilution activity coefficients, the PGL 6th-edition IDAC database for Table 2, and the OPERA compilation³⁴ for the pKa data. Four disclosed defects sit on the deposited record, in §S1, §S8 and §S3: the three-seed surrogate run’s per-run split provenance was not logged, one σ -stream build on disk was written against the wrong split files, the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate, and the COSMO-SAC-2010 layer’s reference reproduction (§2.3) is not deposited, its validator having written no artifact. Two further defects are repaired rather than disclosed, each described where its numbers are printed: the two FastSolv retraining bundles were re-scored (Tables 5 and S17, with the diagnosis, the withdrawn values and the control that reproduces them in §S4) and the withdrawn bundle is kept beside the re-scored one, and the truncated seed-44 σ -oracle per-row file has been recovered complete and now stands in the deposit (Table S25, §S8). What that deposit does and does not make checkable is limitation (xiii). Several of the numbers the verdict rests on are checkable only by re-running the released code, whose input artifacts are larger than the repository carries. The two decomposition sets are therefore deposited as row-level tables (Data Sets S1 and S2), so those sets and their margins can be recomputed with no code at

all; that does not deliver the provenance of the closure evaluations inside them, nor of the trained arms.

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Supporting Information Available

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, ..., and every reference above that carries an S is to it. It contains supplementary methods; the proofs of Lemma 1 and Remark 1 and three further results; per-arm tables and decomposition robustness; the map’s 59 strata in full, including the 44 the estimator cannot bound; further external ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa construction; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. Two machine-readable tables are deposited with it: the 477-row broad IDAC set (Data Set S1) and the 60-row VT-2005-matched set (Data Set S2); their columns are listed in §S8.

References

- (1) Attia, L.; Burns, J. W.; Doyle, P. S.; Green, W. H. Data-driven organic solubility prediction at the limit of aleatoric uncertainty. *Nature Communications* **2025**, *16*, 7497, FASTSOLV model; earlier ChemRxiv 2024 preprint doi:10.26434/chemrxiv-2024-93qp3.
- (2) Vermeire, F. H.; Chung, Y.; Green, W. H.

- Predicting Solubility Limits of Organic Solutes for a Wide Range of Solvents and Temperatures. *Journal of the American Chemical Society* **2022**, *144*, 10785–10797.
- (3) Winter, B.; Winter, C.; Schilling, J.; Bardow, A. A smile is all you need: Predicting limiting activity coefficients from SMILES with natural language processing. *Digital Discovery* **2022**, *1*, 859–869.
 - (4) Sanchez Medina, E. I.; Linke, S.; Stoll, M.; Sundmacher, K. Gibbs–Helmholtz graph neural network: capturing the temperature dependency of activity coefficients at infinite dilution. *Digital Discovery* **2023**, *2*, 781–798.
 - (5) Wang, J.; Ye, J.; Zhang, W. GHGEAT: Gibbs–Helmholtz-Constrained Graph External Attention for Temperature-Dependent Activity Coefficient Prediction at Infinite Dilution. *Journal of Chemical Information and Modeling* **2026**, *66*, 3759–3773.
 - (6) Lin, S.-T.; Sandler, S. I. A Priori Phase Equilibrium Prediction from a Segment Contribution Solvation Model. *Industrial & Engineering Chemistry Research* **2002**, *41*, 899–913.
 - (7) Mullins, E.; Oldland, R.; Liu, Y. A.; Wang, S.; Sandler, S. I.; Chen, C.-C.; Zwolak, M.; Seavey, K. C. Sigma-Profile Database for Using COSMO-Based Thermodynamic Methods. *Industrial & Engineering Chemistry Research* **2006**, *45*, 4389–4415.
 - (8) Yang, Y.; Lin, S.-T. Physics-Embedded Machine Learning Model for Phase Equilibrium Prediction in Multicomponent Systems. *Journal of Chemical Information and Modeling* **2025**, *65*, 10180–10193.
 - (9) Karniadakis, G. E.; Kevrekidis, I. G.; Lu, L.; Perdikaris, P.; Wang, S.; Yang, L. Physics-informed machine learning. *Nature Reviews Physics* **2021**, *3*, 422–440.
 - (10) Raissi, M.; Perdikaris, P.; Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* **2019**, *378*, 686–707.
 - (11) Willard, J.; Jia, X.; Xu, S.; Steinbach, M.; Kumar, V. Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems. *ACM Computing Surveys* **2022**, *55*, 1–37.
 - (12) Ma, L.; Boullé, N.; Yang, Y.-S.; Wu, H.; Guo, L. Physics-Guided Correction for Operator Learning under Model Misspecification. *arXiv preprint arXiv:2606.03469* **2026**,
 - (13) Krishnapriyan, A. S.; Gholami, A.; Zhe, S.; Kirby, R. M.; Mahoney, M. W. Characterizing possible failure modes in physics-informed neural networks. Advances in Neural Information Processing Systems (NeurIPS). 2021.
 - (14) Mohan, A.; Chattopadhyay, A.; Miller, J. What You See is Not What You Get: Neural Partial Differential Equations and The Illusion of Learning. *arXiv preprint arXiv:2411.15101* **2024**,
 - (15) Mahinpei, A.; Clark, J.; Lage, I.; Doshi-Velez, F.; Pan, W. Promises and Pitfalls of Black-Box Concept Learning Models. *arXiv preprint arXiv:2106.13314* **2021**,
 - (16) Margeloiu, A.; Ashman, M.; Bhatt, U.; Chen, Y.; Jamnik, M.; Weller, A. Do Concept Bottleneck Models Learn as Intended? *arXiv preprint arXiv:2105.04289* **2021**,
 - (17) Havasi, M.; Parbhoo, S.; Doshi-Velez, F. Addressing Leakage in Concept Bottleneck Models. Advances in Neural Information Processing Systems (NeurIPS). 2022.
 - (18) Parisini, E.; Chakraborti, T.; Harbron, C.; MacArthur, B. D.; Banerji, C. R. S. Leakage and Interpretability in Concept-Based

Models. *arXiv preprint arXiv:2504.14094* **2025**,

- (19) Espinosa Zarlenga, M.; Dominici, G.; Barbiero, P.; Shams, Z.; Jamnik, M. Avoiding Leakage Poisoning: Concept Interventions Under Distribution Shifts. *arXiv preprint arXiv:2504.17921* **2025**,
- (20) Kennedy, M. C.; O’Hagan, A. Bayesian Calibration of Computer Models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **2001**, *63*, 425–464.
- (21) Brynjarsdóttir, J.; O’Hagan, A. Learning about physical parameters: the importance of model discrepancy. *Inverse Problems* **2014**, *30*, 114007.
- (22) Hirano, K.; Imbens, G. W.; Ridder, G. Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score. *Econometrica* **2003**, *71*, 1161–1189.
- (23) Robins, J. M.; Ritov, Y. Toward a Curse of Dimensionality Appropriate (CODA) Asymptotic Theory for Semi-Parametric Models. *Statistics in Medicine* **1997**, *16*, 285–319.
- (24) Krasnov, L.; Malikov, D.; Kiseleva, M.; Tatarin, S.; Sosnin, S.; Bezzubov, S. Big-SolDB 2.0, dataset of solubility values for organic compounds in different solvents at various temperatures. *Scientific Data* **2025**, *12*, 1236.
- (25) Frenkel, M.; Chirico, R. D.; Diky, V. V.; Yan, X.; Dong, Q.; Muzny, C. ThermoML: An XML-Based Approach for Storage and Exchange of Experimental and Critically Evaluated Thermophysical and Thermochemical Property Data. *Journal of Chemical Information and Computer Sciences* **2003**, *43*, 1344–1359.
- (26) Bell, I. H.; Mickoleit, E.; Hsieh, C.-M.; Lin, S.-T.; Vrabec, J.; Breitzkopf, C.; Jäger, A. A Benchmark Open-Source Implementation of COSMO-SAC. *Journal of Chemical Theory and Computation* **2020**, *16*, 2635–2646.
- (27) Hsieh, C.-M.; Sandler, S. I.; Lin, S.-T. Improvements of COSMO-SAC for Vapor–Liquid and Liquid–Liquid Equilibrium Predictions. *Fluid Phase Equilibria* **2010**, *297*, 90–97.
- (28) Hsieh, C.-M.; Lin, S.-T.; Vrabec, J. Considering the Dispersive Interactions in the COSMO-SAC Model for More Accurate Predictions of Fluid Phase Behavior. *Fluid Phase Equilibria* **2014**, *367*, 109–116.
- (29) de Souza, E. T., Jr.; Alcantara, M. L.; Staudt, P. B.; Coutinho, J. a. A. P.; Soares, R. d. P. Development of a COSMO-SAC Parametrization with Advanced QM Method TZVPD-FINE. *Industrial & Engineering Chemistry Research* **2025**, *64*, 14700–14711.
- (30) Takeishi, N.; Kalousis, A. Physics-Integrated Variational Autoencoders for Robust and Interpretable Generative Modeling. *Advances in Neural Information Processing Systems (NeurIPS)*. 2021.
- (31) Internò, C.; Yamaguchi, J.; Amdahl-Culleton, L.; Olhofer, M.; Klindt, D.; Hammer, B. The Observer Effect in World Models: Invasive Adaptation Corrupts Latent Physics. *arXiv preprint arXiv:2602.12218* **2026**,
- (32) Fingerhut, R.; Chen, W.-L.; Schedemann, A.; Cordes, W.; Rarey, J.; Hsieh, C.-M.; Vrabec, J.; Lin, S.-T. Comprehensive Assessment of COSMO-SAC Models for Predictions of Fluid-Phase Equilibria. *Industrial & Engineering Chemistry Research* **2017**, *56*, 9868–9884.
- (33) Hansch, C.; Leo, A.; Taft, R. W. A Survey of Hammett Substituent Constants and Resonance and Field Parameters. *Chemical Reviews* **1991**, *91*, 165–195.
- (34) Mansouri, K.; Cariello, N. F.; Korotcov, A.; Tkachenko, V.; Grulke, C. M.;

Sprankle, C. S.; Allen, D.; Casey, W. M.; Kleinstreuer, N. C.; Williams, A. J. Open-source QSAR models for pKa prediction using multiple machine learning approaches. *Journal of Cheminformatics* **2019**, *11*, 60.

- (35) Boobier, S.; Hose, D. R. J.; Blacker, A. J.; Nguyen, B. N. Machine learning with physicochemical relationships: solubility prediction in organic solvents and water. *Nature Communications* **2020**, *11*, 5753.

TOC Graphic

